Reorganized Chapter 2: Ramification After Blowup Is Tame

A standalone review copy for comparison with the current chapter

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Chapter 2

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over k. Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i$$

on C, and set $U = C \setminus \mathfrak{m}$.

For each single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, set $d_{\mathfrak{n}} = \deg(\mathfrak{n})$. The effective divisor $\mathfrak{n}$ determines a k-rational closed point

$$Z_{\mathfrak{n}} \hookrightarrow C^{(d_{\mathfrak{n}})} = \mathrm{Sym}_k^{d_{\mathfrak{n}}}(C).$$

Let

$$\pi_{\mathfrak{n}} \colon X_{\mathfrak{n}} := \mathrm{Bl}_{Z_{\mathfrak{n}}}\big(C^{(d_{\mathfrak{n}})}\big) \longrightarrow C^{(d_{\mathfrak{n}})}$$

be the blowup at this point. We denote its exceptional divisor by

$$E_{\mathfrak{n}} := Z_{\mathfrak{n}} \times_{C^{(d_{\mathfrak{n}})}} X_{\mathfrak{n}}$$

By Propositions 22 and 31, $E_{\mathfrak{n}} \cong \mathbb{P}_k^{d_{\mathfrak{n}}-1}$; in particular, it is irreducible and has codimension 1 in $X_{\mathfrak{n}}$. We denote its unique generic point by $\eta_{\mathfrak{n}}$, so that $\overline{\{\eta_{\mathfrak{n}}\}} = E_{\mathfrak{n}}$. Thus we have the Cartesian diagram

$$\begin{tikzcd} E_{\mathfrak{n}} = \overline{\{\eta_{\mathfrak{n}}\}} \arrow[r, hook] \arrow[d] & X_{\mathfrak{n}} \arrow[d, "\pi_{\mathfrak{n}}"] \\ Z_{\mathfrak{n}} \arrow[r, hook] & C^{(d_{\mathfrak{n}})}. \end{tikzcd} \tag{2.1}$$

The purpose of this chapter is to prove Theorem 3. Let $\mathcal{F}$ be the rank-one Λ-local system appearing in that theorem, and let

$$\mathcal{P} = \mathrm{Fr}(\mathcal{F})$$

be its $G = \Lambda^\times$-torsor of frames. By Proposition 9 and definition 26, there is a canonical identification

$$\mathrm{Fr}\big(\mathcal{F}^{[d_{\mathfrak{n}}]}\big) \cong \mathcal{P}^{[d_{\mathfrak{n}}]}.$$

In particular, the two objects have the same ramification. It therefore suffices to prove the following torsor formulation.

Theorem 1. *Let G be a finite abelian group, and let $\mathcal{P} \to U$ be a G-torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, the symmetric-power torsor $\mathcal{P}^{[d_{\mathfrak{n}}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

The chapter is organized around a single observation: the proof of Theorem 1 is essentially the *same computation carried out three times*, once for each of the cyclic building blocks of G:

$$G = \mathbb{Z}/e\mathbb{Z} \;\; (\gcd(e,p) = 1), \qquad G = \mathbb{Z}/p\mathbb{Z}, \qquad G = \mathbb{Z}/p^r\mathbb{Z},$$

governed respectively by Kummer theory, Artin–Schreier theory, and Artin–Schreier–Witt theory. We therefore proceed as follows. Section 2.1 reviews the ramification theory of discrete valuations, and Section 2.2 defines what it means for a G-torsor to be (tamely) ramified, or to have ramification bounded by a modulus, and establishes the basic stability properties of these notions. Section 2.3 then presents the three cyclic theories *in parallel*: each one classifies cyclic extensions by an explicit equation, and reads the ramification off the valuation of a reduced representative; a comparison table summarizes the correspondence. Section 2.4 is a prelude collecting the geometric background propositions shared by the three computations — the local ring at the generic point of the exceptional divisor, the formal–local invariance of the problem, the reduction to $\mathbb{P}^1$, and the invariant-theoretic description of the symmetric-power cover. With these tools in place, Section 2.5 carries out the three computations side by side, following the same three-step skeleton (*realize* the local torsor by an explicit equation over $\mathbb{G}_m$; *identify* the symmetric-power cover at $\eta_{\mathfrak{n}}$; *count* valuations to conclude), and then assembles the general case of Theorem 1 by decomposing G into primary cyclic factors. Finally, Section 2.6 proves auxiliary results on products of blowups used in Chapter 3.

2.1 Ramification of Discrete Valuations

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [Stacks, Tag 0EXQ] and [Ser79].

Definition 2 ([Stacks, Tag 09E4])**.** We say that $A \to B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \to B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B, then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B. The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A. We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K, let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L. Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i. We have

$$[L : K] = \sum\nolimits_{i=1,\ldots,n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 3 ([Stacks, Tag 09E9])**.** Let A be a discrete valuation ring with fraction field K. Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \ldots, n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i,
2. tamely ramified with respect to A if either the characteristic of κ_A is 0 or the characteristic of κ_A is $p > 0$, the field extensions $\kappa(\mathfrak{m}_i)/\kappa_A$ are separable, and the ramification indices e_i are prime to p, and
3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

For the remainder of this section, assume A is a *complete discrete valuation ring* with uniformizer π and a perfect residue field κ. In the equal characteristic case, where $\mathrm{char}(A) = \mathrm{char}(\kappa) = p > 0$, the Cohen Structure Theorem implies that A contains a coefficient field $k \cong \kappa$, such that $A \cong k[[\pi]] \cong k[[t]]$. Consequently, the fraction field is given by $K = k((\pi))$. Let L/K be a finite Galois extension with Galois group G. The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G, denoted by $(G_i)_{i \geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B. In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K. Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields (complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B. Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i.

Consider a tower of fields $K \subset E \subset L$, and let $H = \mathrm{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G, the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_iH/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau \in H} \min(i_K^L(\tau), i+1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \to [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i+1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^iH/H = (G/H)^i$$

for all $i \geq -1$.

We record, for orientation, the structure of the inertia group and of the graded pieces of the filtration; these facts are not used in the sequel, but they explain the dichotomy — prime-to-p versus p-power — that shapes the whole chapter.

Proposition 4 ([Ser79], IV Corollary 3)**.** *The quotients G_i/G_{i+1}, $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p. The group G_1 is a p-group.*

Proposition 5 ([Ser79], IV Corollary 4)**.** *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p.*

2.2 Ramification of G-Torsors

Generally speaking, assume we are given a locally Noetherian scheme X, a dense open $U \subset X$, a finite étale morphism $f : Y \to U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \mathrm{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{tikzcd} \mathrm{Spec}(K_\xi) \arrow[r] \arrow[d] & U \arrow[d] \\ \mathrm{Spec}(\mathcal{O}_{X,\xi}) \arrow[r] & X \end{tikzcd}$$

of schemes. In particular, we see that $Y \times_U \mathrm{Spec}(K_\xi)$ is the spectrum of a finite separable algebra L_ξ/K_ξ.

Definition 6. We say:

1. Y is *unramified* over X at (Z,ξ) resp. Y is *tamely ramified* over X at (Z,ξ) if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ (Definition 3)

2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1 if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ for every (Z,ξ) as above.

More precisely, we decompose L_ξ into a product of finite separable field extensions of K_ξ and we require each of these to be unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$.

Now back to our original assumptions, C is a projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z,ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \to U$ is actually a G-torsor $\mathcal{P} \to U$ in $U_{\text{Ét}}$ for G a finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\mathrm{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction yields a G-torsor over a field, which necessarily decomposes into a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F)$$

such that:

1. These extensions $F/\widehat{K}_P$ are all isomorphic and Galois.

2. The Galois group $H = \mathrm{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of any connected component of the pullback $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$, which is a subgroup of G.

3. The number of such components is then given by the index $[G : H]$.

Definition 7. We say that the extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 8. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. We say the G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if, for each point P_i, the local restriction $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i.

This local property remains invariant under base change to the separable closure. Let $\bar{s} = \mathrm{Spec}(\bar{k}) \to \mathrm{Spec}(k)$ be a geometric point. The higher ramification groups for $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ are isomorphic to those of the geometric restriction $\mathcal{P}_{\bar{k}}$ at any point $\bar{x}$ lying over P. Thus an equivalent definition:

Definition 9. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C. The G-torsor $\mathcal{P} \to U$ has **ramification bounded by** $\mathfrak{m}$ if for every geometric point $\bar{x}$ of $\mathfrak{m}$ (with image $\bar{s}$ in $\mathrm{Spec}(k)$), the restriction of $\mathcal{P}$ to

$$\mathrm{Spec}(\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}}) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{s}}$ at $\bar{x}$ in the sense of Definition 7.

These definitions are equivalent because $\widehat{\mathcal{O}}_{C_{\bar{k}},\bar{x}} \cong \widehat{\mathcal{O}}_{C,P} \otimes_k \bar{k}$. Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in Definition 6.

There is another equivalent formulation through the language of characters. The group G, being a finite abelian group over k, is étale. Consequently, the correspondence between G-torsors and the fundamental group allows us to describe G-torsors via characters. Specifically, there is an isomorphism (Corollary 14):

$$\mathrm{Hom}_{\mathrm{cont.}}(\pi_1^{\text{ét}}(U, \overline{x}), G) \xrightarrow{\cong} \mathrm{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \mathrm{Gal}(\widehat{K}_P^{\mathrm{sep}}/\widehat{K}_P)$. Thus, the restricted G-torsor $\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \to G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r-th ramification group in the upper numbering:

$$\rho(G^r_{\widehat{K}_P}) = \{1\}$$

Basic Properties of Ramification of G-Torsors

We now prove some elementary lemmas regarding the ramification of G-torsors.

Lemma 10. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k. Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. Assume $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G-torsors on $U_{\text{ét}}$, such that $\mathcal{P}_1$ has ramification bounded by r_1 at (Z, ξ), and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z, ξ). Then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by* $\max(r_1, r_2)$ *at (Z, ξ).*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$ and let $K = \mathrm{Frac}(A)$. Let $\rho_1, \rho_2 : G_K \to G$ be the associated continuous homomorphisms corresponding to the G-torsors $\mathcal{P}_1|_{\mathrm{Spec}(K)}$, $\mathcal{P}_2|_{\mathrm{Spec}(K)}$. We finish by noticing that the associated character of $(\mathcal{P}_1 \otimes^G \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = \rho_1 + \rho_2$. □

Lemma 11. *Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$.*

Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors of X and Y such that $Z_X \cap U_X = \emptyset$ and $Z_Y \cap U_Y = \emptyset$. Suppose $f(\xi_X) = \xi_Y$ and that f is étale at ξ_X. Let $\mathcal{P}$ be a G-torsor over U_Y, and let $f^{-1}\mathcal{P}$ be its pullback to $f^{-1}(U_Y) \subset U_X$. Then, the ramification of $f^{-1}\mathcal{P}$ at ξ_X is bounded by r if and only if the ramification of $\mathcal{P}$ at ξ_Y is bounded by r.

Proof. The boundedness of ramification is determined by the behavior of the torsor over the completion of the local rings at the generic points. Let $A = \mathcal{O}_{Y,\xi}$ and $B = \mathcal{O}_{X,\xi_X}$ be the discrete valuation rings at the generic points, with fraction fields K and L respectively. Since f is étale at ξ_X, the map $A \to B$ is a flat, unramified (hence weakly unramified) local homomorphism. Consequently, the extension of completions $\widehat{L}/\widehat{K}$ is a finite unramified extension of complete discretely valued fields. We finish by noticing that the upper numbering filtration on the absolute Galois group is compatible with unramified base change.[1] □

Lemma 12. *Let G_1, G_2 be finite abelian groups and let $X, U, (Z, \xi)$ be as in Lemma 10. Let $\mathcal{P}_1$ be a G_1-torsor and $\mathcal{P}_2$ a G_2-torsor on $U_{ét}$, and consider the fiber product $\mathcal{P}_1 \times_U \mathcal{P}_2$, which is a $(G_1 \times G_2)$-torsor on $U_{ét}$. If $\mathcal{P}_i$ has ramification bounded by r_i at (Z, ξ) for $i = 1, 2$, then $\mathcal{P}_1 \times_U \mathcal{P}_2$ has ramification bounded by $\max(r_1, r_2)$ at (Z, ξ). Moreover, $\mathcal{P}_1 \times_U \mathcal{P}_2$ is tamely ramified (resp. unramified) at (Z, ξ) if and only if both $\mathcal{P}_1$ and $\mathcal{P}_2$ are.*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$, $K = \mathrm{Frac}(A)$, and let $\rho_i : G_K \to G_i$ be the characters of $\mathcal{P}_i|_{\mathrm{Spec}(K)}$. The character of $(\mathcal{P}_1 \times_U \mathcal{P}_2)|_{\mathrm{Spec}(K)}$ is $\rho = (\rho_1, \rho_2) : G_K \to G_1 \times G_2$, since torsors under a product group correspond to pairs of characters, $H^1(K, G_1 \times G_2) = H^1(K, G_1) \times H^1(K, G_2)$, compatibly with the projections. Now for any closed subgroup $N \subset G_K$ (we take the upper-numbering group G_K^r, resp. the wild inertia subgroup, resp. the inertia subgroup) we have $\rho(N) = \{1\}$ if and only if $\rho_1(N) = \{1\}$ and $\rho_2(N) = \{1\}$. □

Lemma 13. *Let $\iota : H \hookrightarrow G$ be an injective homomorphism of finite abelian groups, let $X, U, (Z, \xi)$ be as above, let $\mathcal{Q}$ be an H-torsor on $U_{ét}$, and let $\iota_*\mathcal{Q} := (\mathcal{Q} \times G)/H$ be the induced G-torsor (where H acts by $h \cdot (q, g) = (qh,\ \iota(h)^{-1}g)$). Then $\iota_*\mathcal{Q}$ has ramification bounded by r at (Z, ξ) if and only if $\mathcal{Q}$ does; likewise for tame ramification and for unramifiedness.*

Proof. Let $A = \widehat{\mathcal{O}_{X,\xi}}$, $K = \mathrm{Frac}(A)$, and let $\chi : G_K \to H$ be the character of $\mathcal{Q}|_{\mathrm{Spec}(K)}$; the character of $(\iota_*\mathcal{Q})|_{\mathrm{Spec}(K)}$ is $\iota \circ \chi$. Since ι is injective, $\iota \circ \chi$ and χ have the same kernel, so for any closed subgroup $N \subset G_K$ we have $(\iota \circ \chi)(N) = \{1\}$ if and only if $\chi(N) = \{1\}$. (Concretely: the connected components of the two torsors over $\mathrm{Spec}(K)$ are spectra of the *same* field extension F/K, the fixed field of $\ker \chi$; the induced torsor merely has $[G : H]$ times as many components.) □

2.3 Kummer, Artin–Schreier, and Artin–Schreier–Witt Theories

For cyclic extensions, Kummer, Artin–Schreier, and Artin–Schreier–Witt theories provide explicit ways to classify extensions and calculate their ramification. The three theories follow one and the same template. In each case there is a short exact sequence of commutative group schemes over the base field K,

$$1 \to \mu_e \to \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \to 1, \qquad 0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \to 0, \qquad 0 \to \mathbb{Z}/p^r\mathbb{Z} \to W_r \xrightarrow{\wp} W_r \to 0,$$

whose long exact sequence in (étale or Galois) cohomology classifies the G-torsors over K — for $G = \mathbb{Z}/e\mathbb{Z}$ (with $\mu_e \cong \mathbb{Z}/e\mathbb{Z}$ after a choice of primitive root of unity, requiring $\gcd(e, \mathrm{char}\, K) = 1$ and $\mu_e \subset K$), $G = \mathbb{Z}/p\mathbb{Z}$, and $G = \mathbb{Z}/p^r\mathbb{Z}$ respectively — by an explicit quotient group:

$$K^\times/(K^\times)^e, \qquad K/\wp(K), \qquad W_r(K)/\wp(W_r(K)).$$

[1] Specifically, let $G_K = \mathrm{Gal}(K^{sep}/K)$ and $G_L = \mathrm{Gal}(L^{sep}/L)$. For an unramified extension, the Herbrand function is the identity, which implies that for any $r \geq 0$:

$$G_L^r = G_K^r \cap G_L$$

Concretely, each class is realized by adjoining a solution of one explicit equation: $X^e = a$, resp. $\wp(X) = X^p - X = a$, resp. $\wp(\vec{X}) = \vec{f}$ in Witt coordinates. When K is moreover a local field $k((x))$, each class admits a *reduced representative* — a Laurent polynomial (or vector of Laurent polynomials) in x^{-1} normalized so that the ramification of the extension can be read off directly from the valuation: the ramification index in the Kummer case, and the (highest) upper-numbering break in the Artin–Schreier and Artin–Schreier–Witt cases. The comparison table at the end of the section summarizes the three columns of this dictionary; the computations of Section 2.5 will traverse each column in turn.

Throughout this section, K is a discrete valuation field with a perfect residue field κ of characteristic $p > 0$; from Lemma 15 on, we further specialize to $K = k((x))$ with k algebraically closed.

2.3.1 Kummer Theory

Let e be a positive integer prime to p, and assume K contains the group μ_e of e-th roots of unity (of order e, since $\gcd(e, p) = 1$ and κ is perfect). Kummer theory states that every cyclic extension of K of degree dividing e is generated by an e-th root: it is of the form $K(a^{1/e})$ for some $a \in K^\times$, and the assignment $a \mapsto K(a^{1/e})$ induces a bijection between the cyclic subgroups of $K^\times/(K^\times)^e$ and the cyclic extensions of K of degree dividing e, with $\mathrm{Gal}(K(a^{1/e})/K)$ cyclic of order equal to the order of a in $K^\times/(K^\times)^e$, acting by multiplication by roots of unity on $a^{1/e}$. The ramification of these extensions is computed by the valuation of a:

Theorem 14 (Ramification in Kummer Extensions, [Koc97], Proposition 1.83)**.** *Assume K contains the n-th roots of unity μ_n, with $\gcd(n, p) = 1$. Let L/K be the extension given by the equation $X^n = a$ for some $a \in K^\times$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is an n-th power, the extension L/K is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is not an n-th power, the extension L/K is cyclic and unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the extension L/K is cyclic and ramified. Specifically, if $\gcd(|v_K(a)|, n) = 1$, the extension is totally ramified of degree n. Otherwise it has ramification index $\frac{n}{\gcd(|v_K(a)|, n)}$.*

Conversely, Kummer theory ensures that every cyclic extension of degree n, prime to p, of a field that contains the n-th roots of unity, is of the above form. Moreover, in the above we can always take $a \in \mathcal{O}_K$.

Note that in the case of total ramification the extension is tamely ramified; indeed, since $\gcd(n, p) = 1$, *every* extension appearing in Theorem 14 is tamely ramified.

Over the local field $k((x))$ with k algebraically closed, the classes admit a particularly simple reduced representative — a power of the uniformizer:

Lemma 15 (reduced representatives, Kummer case)**.** *Let k be algebraically closed of characteristic $p > 0$, let $K = k((x))$, and let e be a positive integer prime to p. Then:*

1. *Every tamely ramified extension of K is of the form $k((x^{1/e'}))$ for some e' prime to p; consequently the tame quotient G^t of $\mathrm{Gal}(K^{sep}/K)$ is procyclic, $G^t \cong \varprojlim_{p\nmid e'} \mu_{e'}(k) \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$.*

2. *Every class in $K^\times/(K^\times)^e$ is represented by x^a for a unique $a \in \{0,1,\dots,e-1\}$; the classes of the x^a exhaust $\mathrm{Hom}_{\mathrm{cont.}}(G^t, \mathbb{Z}/e\mathbb{Z}) \cong \mathbb{Z}/e\mathbb{Z}$, and the class of x^a is a surjective character (i.e., the extension $X^e = x^a$ is a field, totally ramified of degree e) precisely when $\gcd(a,e) = 1$.*

Proof. (1) Since k is algebraically closed, the residue field of K is algebraically closed, so K admits no non-trivial unramified extensions, and its tamely ramified extensions are exactly the Kummer extensions $k((x^{1/e'}))$ with $p \nmid e'$ (Theorem 14). The description of G^t follows by passing to the limit.

(2) Since $k^\times$ is divisible and every unit of $k[[x]]$ with constant term 1 is an e-th power (Hensel; $\gcd(e,p) = 1$), we have $K^\times = x^{\mathbb{Z}} \cdot (K^\times)^e \cdot k^\times = x^{\mathbb{Z}}(K^\times)^e$, so the classes of $x^0, \dots, x^{e-1}$ exhaust $K^\times/(K^\times)^e \cong \mathbb{Z}/e\mathbb{Z}$. Under the Kummer isomorphism $K^\times/(K^\times)^e \cong \mathrm{Hom}_{\mathrm{cont.}}(G^t, \mathbb{Z}/e\mathbb{Z})$ the class of x^a is a times a generator, hence surjective iff $\gcd(a,e) = 1$; by Theorem 14(3) this is exactly the totally ramified case of degree e. ☐

2.3.2 Artin–Schreier Theory

For cyclic extensions of degree $p = \operatorname{char} \kappa$, the role of the e-th power map is played by the *Artin–Schreier operator* $\wp(X) = X^p - X$, and the role of $K^\times/(K^\times)^e$ by the additive quotient $K/\wp(K)$.

Theorem 16 (Ramification in Artin–Schreier Extensions, [Tho05])**.** *Let $\wp(x) = x^p - x$ be the Artin–Schreier operator. Let L/K be the extension given by the equation $X^p - X = a$ for some $a \in K$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) > 0$ or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension L/K is trivial.*
2. *If $v_K(a) = 0$ and if $a \notin \wp(K)$, the extension L/K is cyclic of degree p and unramified.*
3. *If $v_K(a) = -m < 0$ with $m \in \mathbb{Z}_{>0}$ and if m is prime to p, the extension L/K is cyclic of degree p again and totally ramified. Moreover, its ramification groups are given by:*

$$G = G^{(-1)} = \cdots = G^{(m)} \quad \text{and} \quad G^{(m+1)} = 1.$$

Conversely, Artin–Schreier theory ensures that every cyclic extension of degree p takes this form.

The analogue of the representatives x^a of Lemma 15 is a polynomial in x^{-1} whose pole order is prime to p; the pole order is then exactly the unique ramification break of Theorem 16(3).

Lemma 17 (reduced representatives, Artin–Schreier case)**.** *Let k be algebraically closed of characteristic $p > 0$ and let $K = k((x))$. Every class in $K/\wp(K)$ admits a representative $f \in x^{-1}k[x^{-1}]$ which is either 0 or of the form*

$$f = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots + a_{-1}x^{-1}, \qquad c \neq 0, \quad p \nmid m.$$

In the latter case the extension $X^p - X = f$ is cyclic of degree p, totally ramified, with unique ramification break m.

Proof. We show every class admits such a representative:

- $\wp$ is surjective on $k[[x]]$: given $h = \sum_{i \geq 0} b_i x^i$, solve $\wp(u) = h$ with $u = \sum_{i \geq 0} u_i x^i$ by choosing u_0 with $u_0^p - u_0 = b_0$ (possible as k is algebraically closed) and setting recursively $u_i = u_{i/p}^p - b_i$, with the convention $u_{i/p} = 0$ when $p \nmid i$. Hence, writing an arbitrary element as $f^- + f^+$ with $f^- \in x^{-1}k[x^{-1}]$ and $f^+ \in k[[x]]$, we may choose a representative in $x^{-1}k[x^{-1}]$.

- If the representative has pole order m divisible by p, with leading term cx^{-m}, then subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order while staying inside $x^{-1}k[x^{-1}]$. Iterating, we reach a representative f which is either 0 or has pole order m prime to p.

The last assertion is Theorem 16(3), noting that a non-zero reduced f has $v_K(f) = -m < 0$ with $p \nmid m$. □

Cyclic extensions of degree p^r are classified by the same picture with the additive group $\mathbb{G}_a$ replaced by the ring scheme W_r of Witt vectors of length r, and $\wp = F - \mathrm{id}$ the Artin–Schreier–Witt operator. The following subsection collects the required background on Witt vectors and the Artin–Schreier–Witt classification (Theorem 27) together with the Schmid–Witt formula (Theorem 31) computing the ramification breaks — the degree-p^r row of the dictionary.

2.3.3 Witt Vectors and Artin–Schreier–Witt Theory

Artin–Schreier theory, together with the ramification formula of Theorem 16, gives complete control over the cyclic extensions of degree p of a local field of characteristic p. To treat the groups $G = \mathbb{Z}/p^r\mathbb{Z}$ we need the analogous control over cyclic extensions of degree p^r, which is provided by Witt's generalization: the field K is replaced by the ring $W_r(K)$ of p-typical Witt vectors of length r, the operator $\wp(u) = u^p - u$ by its Witt-vector analogue $\wp = F - \mathrm{id}$, and the classical ramification-break formula by the Schmid–Witt formula. This subsection collects the definitions and the elementary lemmas that we use; our references for the general theory are [Ser79, II, §6] and [Rab14].

Throughout, p is a fixed prime, and $r \geq 1$ is an integer. All the Witt vectors appearing in this thesis are p-typical and of finite length.

The ring of Witt vectors

For $n \geq 0$, the n-th **Witt polynomial** in the variables $X_0, \ldots, X_n$ is

$$w_n(X_0, \ldots, X_n) \;=\; \sum_{j=0}^{n} p^j X_j^{p^{n-j}} \;=\; X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Theorem 18 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1])**.** *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \ldots, X_n; Y_0, \ldots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \ldots, S_n) = w_n(X_0, \ldots, X_n) + w_n(Y_0, \ldots, Y_n), \qquad w_n(P_0, \ldots, P_n) = w_n(X_0, \ldots, X_n) \cdot w_n(Y_0, \ldots, Y_n)$$

for all n. For any commutative ring A, the operations

$$\vec{a} \oplus \vec{b} := \big(S_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}, \qquad \vec{a} \odot \vec{b} := \big(P_n(\vec{a}; \vec{b})\big)_{0 \leq n \leq r-1}$$

*make the set A^r into a commutative ring $W_r(A)$, the ring of **p-typical Witt vectors of length r over** A, with zero $(0, \ldots, 0)$ and unit $(1, 0, \ldots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map***

$$w \colon W_r(A) \longrightarrow A^r, \qquad \vec{a} \longmapsto \big(w_0(\vec{a}), w_1(\vec{a}), \ldots, w_{r-1}(\vec{a})\big)$$

is a ring homomorphism to the product ring A^r; it is injective when p is a non-zero-divisor in A, and bijective when $p \in A^\times$.

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma\colon A \to B$ induces the ring homomorphism

$$W_r(\sigma)\colon W_r(A) \to W_r(B), \qquad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ-invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma \;=\; W_r\big(A^\Gamma\big) \qquad \text{inside } W_r(A). \tag{2.2}$$

The uniqueness statement in Theorem 18 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 19 (ghost principle)**.** *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by Theorem 18 and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \to A$. ☐

The only structural feature of the addition polynomials S_n that we shall need is their "triangular" shape:

Proposition 20 (shape of Witt addition)**.** *For every $n \geq 0$ there is a polynomial $C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-}$ involving only the variables of index* strictly less *than n, such that*

$$(\vec{a} \oplus \vec{b})_n \;=\; a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \tag{2.3}$$

We refer to C_n as the ***carry polynomial****. The analogous statement holds for $\ominus$.*

Proof. By induction on n, using the recursion defining S_n:

$$p^n S_n \;=\; \sum_{j \leq n} p^j \big(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\big) \;-\; \sum_{j<n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients (Theorem 18), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. ☐

Frobenius, Verschiebung and truncation

From now on, A denotes an $\mathbb{F}_p$-algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 21. Let A be an $\mathbb{F}_p$-algebra.

1. The **Frobenius** $F\colon W_r(A) \to W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A\colon a \mapsto a^p$ is the absolute Frobenius of A. Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.

2. The **Verschiebung** is the shift
$$V\colon W_{r-1}(A) \to W_r(A), \qquad V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}).$$
Iterating, $V^j\colon W_{r-j}(A) \to W_r(A)$; in particular $V^{r-1}\colon A = W_1(A) \to W_r(A)$ sends $c \mapsto (0, \dots, 0, c)$.

3. The **truncation** is
$$t\colon W_r(A) \to W_{r-1}(A), \qquad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$-) algebra A, the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 22. *Let A be an $\mathbb{F}_p$-algebra. Then:*

1. *t is a natural ring homomorphism commuting with F, and its kernel is*
$$\ker\bigl(t\colon W_r(A) \to W_{r-1}(A)\bigr) \;=\; V^{r-1}A \;=\; \{(0, \dots, 0, c) : c \in A\}.$$

2. *V is additive, and $FV = VF$.*

3. *Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,*
$$p \cdot (a_0, \dots, a_{r-1}) \;=\; (0,\, a_0^p,\, \dots,\, a_{r-2}^p).$$
Iterating, $p^j \cdot \vec{a} = (0, \dots, 0, a_0^{p^j}, \dots, a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and
$$p^{r-1} \cdot (a_0, \dots, a_{r-1}) \;=\; \bigl(0, \dots, 0,\, a_0^{p^{r-1}}\bigr). \tag{2.4}$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n: the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 20). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 19 and the ghost identity $w_n(V\vec{a}) = p\, w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate since F is componentwise. (3) The identity $FV = p$ holds over arbitrary rings for the general Frobenius, by Lemma 19 applied to $w_n(FV\vec{a}) = w_{n+1}(V\vec{a}) = p\, w_n(\vec{a}) = w_n(p\vec{a})$; in characteristic p the general Frobenius is componentwise, giving the displayed formula. See [Ser79, II, §6, Proposition 8 ff.] for details. ∎

Example 23. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (2.4) the unit $\underline{1} = (1, 0, \ldots, 0)$ satisfies $p^j \cdot \underline{1} = (0, \ldots, 0, 1, 0, \ldots, 0)$ (1 in position j), so $\underline{1}$ has additive order p^r; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

Definition 24. The **Artin–Schreier–Witt operator** on $W_r(A)$, A an $\mathbb{F}_p$-algebra, is

$$\wp \;:=\; F - \mathrm{id}, \qquad \wp\vec{u} \;=\; F\vec{u} \ominus \vec{u}.$$

For $r = 1$ this is the classical operator $\wp(u) = u^p - u$ of Theorem 16.

Since F is a ring endomorphism, $\wp$ is an endomorphism of the additive group $(W_r(A), \oplus)$: $\wp(\vec{u} \oplus \vec{v}) = \wp\vec{u} \oplus \wp\vec{v}$. Moreover $\wp$ commutes with $W_r(\sigma)$ for every ring homomorphism σ (both F and the ring operations are natural), and with V and t by Proposition 22.

The following bookkeeping lemma is elementary but does the essential work of reducing statements about W_r to statements about W_{r-1} and about A itself: the truncation t and the inclusion V^{r-1} of its kernel slice W_r into lower-length pieces, compatibly with $\wp$, and *addition of an element of the kernel is carry-free.*

Lemma 25 (Witt bookkeeping)**.** *Let A be an $\mathbb{F}_p$-algebra.*

(i) $t\colon W_r(A) \to W_{r-1}(A)$ is a ring homomorphism commuting with F, hence with $\wp$; its kernel is $V^{r-1}A$.

(ii) $V^{r-1}\colon A \to W_r(A)$ is additive, and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$, where on the left $\wp$ is the operator on $W_r(A)$ and on the right the classical $\wp(c) = c^p - c$ on A.

(iii) (carry-free top-level addition) For all $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) \;=\; (a_0, \ldots, a_{r-2},\, a_{r-1} + c).$$

Proof. (i) and (ii) restate parts of Proposition 22 (for (ii): V^{r-1} is additive and commutes with the componentwise F, hence with $\wp = F - \mathrm{id}$). (iii) By Lemma 19 it suffices to verify the identity over $\mathbb{Z}[a_0, \ldots, a_{r-1}, c]$, where the ghost map is injective. Put $\vec{b} := (0, \ldots, 0, c)$ and $\vec{s} := \vec{a} \oplus \vec{b}$. For $n < r - 1$ we have $w_n(\vec{b}) = 0$, so $w_n(\vec{s}) = w_n(\vec{a})$, and by induction on n (solving for s_n, using that p is a non-zero-divisor) $s_n = a_n$ for $n \le r - 2$. For $n = r - 1$, $w_{r-1}(\vec{b}) = p^{r-1}c$ gives $p^{r-1}s_{r-1} = p^{r-1}a_{r-1} + p^{r-1}c$, i.e. $s_{r-1} = a_{r-1} + c$. □

Remark. Lemma 25(iii) fails for the addition of vectors supported in lower positions: in general the carry polynomials C_n of (2.3) are non-zero, and Witt addition is *not* componentwise. In particular, operations performed componentwise on a Witt vector — such as discarding the regular parts of Laurent-series components — do *not* in general respect Witt addition, and will be carefully avoided.

The Artin–Schreier–Witt isogeny and its torsors

The componentwise ring operations make $\mathbb{A}^r_{\mathbb{F}_p}$ a ring scheme, the **Witt vector group scheme** $\mathbb{W}_r$: its underlying scheme is $\mathbb{A}^r_{\mathbb{F}_p} = \operatorname{Spec} \mathbb{F}_p[T_0, \ldots, T_{r-1}]$, with the group law given by the addition polynomials S_n, so that $\mathbb{W}_r(A) = W_r(A)$ functorially. The operator $\wp = F - \mathrm{id}$ is an endomorphism of the group scheme $\mathbb{W}_r$, and it is the Witt-vector instance of a *Lang isogeny*: by Example 23 and the lemma below, it sits in a short exact sequence of group schemes over $\mathbb{F}_p$ for the étale topology

$$0 \longrightarrow \mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \tag{2.5}$$

The corresponding covers admit the following completely explicit description, which is the form in which we use them. For $\vec{g} \in W_r(A)$ we write $A[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$, $\vec{X} = (X_0, \ldots, X_{r-1})$, for the quotient of $A[X_0, \ldots, X_{r-1}]$ by the ideal generated by the r components of the Witt vector $\wp\vec{X} \ominus \vec{g} \in W_r(A[X_0, \ldots, X_{r-1}])$.

Lemma 26 (ASW covers are étale torsors)**.** *Let A be an $\mathbb{F}_p$-algebra and $\vec{g} \in W_r(A)$. Let*

$$B \;:=\; A[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big).$$

Then B is finite free over A with basis $\big\{\prod_n X_n^{a_n} : 0 \leq a_n \leq p-1\big\}$, and étale over A. The translation action of $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$,

$$m\colon \vec{X} \longmapsto \vec{X} \oplus \underline{m},$$

makes $\operatorname{Spec} B$ *a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over* $\operatorname{Spec} A$.

Proof. By (2.3) and the componentwise formula for F, the n-th defining equation has the triangular form

$$X_n^p - X_n - g_n + C'_n(X_0, \ldots, X_{n-1}; g_0, \ldots, g_{n-1}) \;=\; 0$$

for universal integral polynomials C'_n. Triangularity gives the free basis; the Jacobian matrix of the system is lower triangular with diagonal entries $pX_n^{p-1} - 1 = -1$, a unit, so B is étale over A. Finally, $\operatorname{Spec} B$ is the pullback along $\vec{g}\colon \operatorname{Spec} A \to \mathbb{W}_r$ of the covering $\wp\colon \mathbb{W}_r \to \mathbb{W}_r$; by the triangular form just established (over $A = \mathbb{F}_p[T_0, \ldots, T_{r-1}]$, $\vec{g} = (T_0, \ldots, T_{r-1})$), $\wp$ is a surjective étale homomorphism of group schemes, and its kernel is $\{\vec{u} : F\vec{u} = \vec{u}\} = W_r(\mathbb{F}_p)$ since F is componentwise. A surjective étale group homomorphism is a torsor under its kernel, and torsors pull back to torsors; hence $\operatorname{Spec} B$ is a $W_r(\mathbb{F}_p)$-torsor over $\operatorname{Spec} A$. □

Artin–Schreier–Witt theory for fields

Over a field, the sequence (2.5) computes all cyclic extensions of p-power degree, exactly as classical Artin–Schreier theory does for degree p.

Theorem 27 (Artin–Schreier–Witt theory; [Tho05, §3–4])**.** *Let $M \supseteq \mathbb{F}_p$ be a field, M^s a separable closure, and $\Gamma = \operatorname{Gal}(M^s/M)$. Then $\wp$ is surjective on $W_r(M^s)$ with kernel $W_r(\mathbb{F}_p)$, and there is a canonical isomorphism of abelian groups*

$$\delta\colon\; W_r(M)/\wp W_r(M) \;\xrightarrow{\cong}\; \operatorname{Hom}_{\mathrm{cont}}\big(\Gamma,\, \mathbb{Z}/p^r\mathbb{Z}\big) \;=\; H^1\big(M, \mathbb{Z}/p^r\mathbb{Z}\big), \qquad [\vec{g}] \longmapsto \big(\sigma \mapsto \sigma(\vec{u}) \ominus \vec{u}\big),$$

where $\vec{u} \in W_r(M^s)$ is any solution of $\wp\vec{u} = \vec{g}$. Under the correspondence between characters and torsors (Corollary 14), the class $[\vec{g}]$ corresponds to the $\mathbb{Z}/p^r\mathbb{Z}$-torsor

$$\operatorname{Spec}\, M[\vec{X}]/\big(\wp\vec{X} \ominus \vec{g}\big)$$

of Lemma 26. Moreover, for a class of order p^s in $W_r(M)/\wp W_r(M)$, the fixed field $M(\vec{u}) := M(u_0, \ldots, u_{r-1})$ of $\ker \delta[\vec{g}]$ is a cyclic extension of M of degree p^s; the torsor is connected if and only if $[\vec{g}]$ has order p^r, in which case $\operatorname{Spec} M[\vec{X}]/(\wp\vec{X} \ominus \vec{g}) \cong \operatorname{Spec} M(\vec{u})$ is (the spectrum of) a cyclic field extension of degree p^r. Every cyclic extension of M of degree dividing p^r arises this way.

Sketch of proof. Surjectivity of $\wp$ on $W_r(M^s)$ and the computation of its kernel follow from the triangular equations in the proof of Lemma 26: each component of a preimage is obtained by solving a separable equation $X_n^p - X_n = (\dots)$, and $F\vec{u} = \vec{u}$ holds if and only if $u_n^p = u_n$ for all n, i.e. $\vec{u} \in W_r(\mathbb{F}_p)$. Given $\vec{g}$ and a solution $\wp\vec{u} = \vec{g}$, for $\sigma \in \Gamma$ we have $\wp(\sigma\vec{u} \ominus \vec{u}) = \sigma\vec{g} \ominus \vec{g} = 0$ (using (2.2) and that $W_r(\sigma)$ commutes with $\wp$), so $\sigma\vec{u} \ominus \vec{u} \in W_r(\mathbb{F}_p)$; one checks it is a continuous homomorphism in σ (the kernel is fixed pointwise by Γ), independent of the choices modulo nothing and of the representative $\vec{g}$ modulo $\wp W_r(M)$. Injectivity: if $\delta[\vec{g}] = 0$ then $\vec{u}$ is Γ-invariant, hence lies in $W_r(M)$ by (2.2), so $\vec{g} = \wp\vec{u} \in \wp W_r(M)$. Surjectivity of δ is the additive Hilbert 90: $H^1(\Gamma, W_r(M^s)) = 0$, by induction on r along the exact sequence $0 \to M^s \xrightarrow{V^{r-1}} W_r(M^s) \xrightarrow{t} W_{r-1}(M^s) \to 0$ of Γ-modules (Lemma 25(i)) and the classical case $H^1(\Gamma, M^s) = 0$. The remaining statements follow from Theorem 13 applied to the character $\delta[\vec{g}]$, whose image has the same order as the class $[\vec{g}]$. For details see [Tho05, §3–4] or [Rab14]. ☐

The order of a class is partially visible in its 0-th component:

Corollary 28. *Let $M \supseteq \mathbb{F}_p$ be a field and $\vec{g} \in W_r(M)$ with $g_0 = 0$. Then $p^{r-1}[\vec{g}] = 0$ in $W_r(M)/\wp W_r(M)$; equivalently, the cyclic extension attached to $[\vec{g}]$ has degree at most p^{r-1}. Consequently, if $[\vec{g}]$ has order p^r (e.g. if $\vec{g}$ defines a connected torsor), then* every *representative of the class has non-zero 0-th component.*

Proof. By (2.4), $p^{r-1}\vec{g} = (0, \dots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. ☐

Remark. When $M = k((x))$ with k algebraically closed, every finite separable extension L/M is totally ramified: the residue field extension is trivial (as k is algebraically closed) and $\mathcal{O}_L$ is a complete discrete valuation ring, so $[L : M] = ef = e$. Thus a class of order p^r in $W_r(k((x)))/\wp$ corresponds to a cyclic, *totally ramified* extension of degree p^r.

Isobaricity of the Witt addition polynomials

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (2.3) are complicated, they are *isobaric* for the weighting in which the j-th component has weight p^j.

Lemma 29 (isobaricity of Witt addition)**.** *Assign to the variables X_j and Y_j the weight p^j. Then the n-th addition polynomial S_n is isobaric of weight p^n: every monomial of S_n has total weight exactly p^n. The same holds for each component of a d-fold iterated Witt sum $\bigoplus_{i=1}^d \vec{a}_i$, viewed as a universal polynomial in the components $a_{i,j}$ of weight p^j.*

Proof. The n-th ghost polynomial $w_n = \sum_{j\le n} p^j X_j^{p^{n-j}}$ is isobaric of weight p^n for this weighting. The recursion

$$p^n S_n = \sum_{j\le n} p^j \left(X_j^{p^{n-j}} + Y_j^{p^{n-j}}\right) - \sum_{j<n} p^j S_j^{p^{n-j}}$$

preserves isobaricity of weight p^n by induction on n: each S_j is isobaric of weight p^j, so $S_j^{p^{n-j}}$ is isobaric of weight p^n. Isobaricity persists under iterated substitution of Witt sums into Witt sums (substituting an isobaric polynomial of weight p^j for a variable of weight p^j preserves weights), giving the statement for d-fold sums. ☐

Ramification breaks: the Schmid–Witt formula

Finally we state the generalization of the ramification formula of Theorem 16 to Artin–Schreier–Witt extensions. As in the classical case, the formula reads off the largest upper-numbering ramification break from the pole orders of a defining Witt vector — provided the representative is normalized so that no interference between the levels can occur. The relevant normalization is the following.

Definition 30. Let $K = k((x))$. For $g \in K$ put $m(g) := \max\bigl(0, -v_x(g)\bigr)$ (the pole order of g, or 0 if g is regular). A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$ is **reduced** if for every j, either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Theorem 31 (Schmid–Witt break formula)**.** *Let $K = k((x))$ with k perfect of characteristic p, and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class (Theorem 27) is represented by a* reduced *vector $\vec{g} \in W_r(K)$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K equals*

$$u \;=\; \max_{0 \le j \le r-1} \; p^{\,r-1-j}\, m_j.$$

This is due, for finite residue fields, to Kanesaka–Sekiguchi [KS79] and Thomas [Tho05] (building on Schmid [Sch37] for $r = 1$), and for arbitrary perfect residue fields to Elder–Keating [EK25]. We use it as stated, in one direction only: *evaluating the formula on one reduced representative computes u*; no minimization over representatives is needed.

Remark. Reducedness is exactly what makes the formula exact. If two levels $j < j'$ with $m_j, m_{j'} \neq 0$ satisfied $p^{r-1-j} m_j = p^{r-1-j'} m_{j'}$, then $m_{j'} = p^{j'-j} m_j$ would be divisible by p, contradicting Definition 30; hence the maximum is attained at a unique level, and no cancellation of breaks between levels can occur.

Remark. For $r = 1$, a reduced vector is a single element $g \in K$ whose pole order m_0 is zero or prime to p, and Theorem 31 recovers Theorem 16: the largest (indeed the only) upper break is m_0. The existence of reduced representatives — and, over $K = k((x))$, of reduced representatives whose components are polynomials in x^{-1} without constant term — is not part of the general theory quoted here; it will be proved where it is used, by induction on r through the truncation maps of Lemma 25.

2.3.4 Reduced Representatives for Artin–Schreier–Witt Classes

We now record the W_r-analogue of Lemma 17: every Artin–Schreier–Witt class over $k((x))$ is represented by a vector of *polynomials in x^{-1}* with controlled pole orders. Recall from Definition 30 the notion of a reduced vector; we add:

Definition 32. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced (Definition 30) and moreover every $g_j \in x^{-1}k[x^{-1}]$; then $m(g_j) = \deg_{x^{-1}} g_j$, and $m(g_j) = 0$ if and only if $g_j = 0$.

Lemma 33 (surjectivity of $\wp$ on integral Witt vectors)**.** *$\wp\colon W_r(k[[x]]) \to W_r(k[[x]])$ is surjective.*

Proof. Induction on r. For $r = 1$ this is the first bullet in the proof of Lemma 17: given $h = \sum_{i \ge 0} b_i x^i$, solve $u^p - u = h$ coefficientwise, using that k is algebraically closed for the constant term. For $r > 1$, let $\vec{h} \in W_r(k[[x]])$. By induction choose $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp\vec{u}' = t(\vec{h})$, and lift it to $\tilde{u} := (\vec{u}', 0) \in W_r(k[[x]])$. By Lemma 25(i), $t\bigl(\wp\tilde{u} \ominus \vec{h}\bigr) = \wp\vec{u}' \ominus t(\vec{h}) = 0$, so $\wp\tilde{u} \ominus \vec{h} = V^{r-1}(g)$

for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$ (case $r = 1$ again). Then by Lemma 25(ii) and additivity of V^{r-1},

$$\wp\big(\tilde{u} \oplus V^{r-1}(w)\big) = \wp\tilde{u} \oplus V^{r-1}(\wp w) = \big(\vec{h} \oplus V^{r-1}(g)\big) \oplus V^{r-1}(-g) = \vec{h}. \qquad \square$$

Proposition 34 (reduced polynomial representatives)**.** *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative: for every $\vec{f} \in W_r(k((x)))$ there exists $\vec{u} \in W_r(k((x)))$ such that $\vec{g} := \vec{f} \ominus \wp\vec{u}$ has all components in $x^{-1}k[x^{-1}]$ with pole orders either 0 or prime to p.*

Proof. Induction on r.

Base case $r = 1$. This is Lemma 17: first, writing $f = f^- + f^+$ with $f^- \in x^{-1}k[x^{-1}]$ and $f^+ \in k[[x]]$, surjectivity of $\wp$ on $k[[x]]$ removes f^+; then, as long as the leading term is cx^{-m} with $p \mid m$, $m > 0$, subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order while staying inside $x^{-1}k[x^{-1}]$; this terminates at 0 or at pole order prime to p.

Inductive step. Let $\vec{f} \in W_r(k((x)))$. By induction there is $\vec{u}' \in W_{r-1}(k((x)))$ such that $\vec{g}' := t(\vec{f}) \ominus \wp\vec{u}'$ is a reduced polynomial vector in W_{r-1}. Lift $\vec{u}'$ to $\tilde{u} := (\vec{u}', 0) \in W_r(k((x)))$ and set $\vec{h} := \vec{f} \ominus \wp\tilde{u}$. By Lemma 25(i), $t(\vec{h}) = \vec{g}'$, i.e.

$$\vec{h} = (g'_0, \ldots, g'_{r-2}, h_{r-1}), \qquad h_{r-1} \in k((x)),$$

with the first $r-1$ components already reduced polynomials. It remains to repair the last component *without touching the others*; by Lemma 25(iii), adjustments of the form $\ominus\,\wp\big(V^{r-1}(z)\big) = \ominus\, V^{r-1}(\wp z)$ change only the last component, by $-\wp(z)$. So:

- Write $h_{r-1} = h^- + h^+$ with $h^- \in x^{-1}k[x^{-1}]$, $h^+ \in k[[x]]$, choose $w \in k[[x]]$ with $\wp w = h^+$ (Lemma 33 with $r = 1$), and replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}w)$: the last component becomes h^-.
- While the last component has leading term cx^{-m} with $p \mid m$, $m > 0$, replace $\vec{h}$ by $\vec{h} \ominus \wp\big(V^{r-1}(c^{1/p}x^{-m/p})\big)$: the last component changes by $-\wp(c^{1/p}x^{-m/p}) = -cx^{-m} + c^{1/p}x^{-m/p}$, so its pole order strictly decreases and it stays in $x^{-1}k[x^{-1}]$. This terminates at 0 or at pole order prime to p.

The result is a reduced polynomial vector $\wp$-equivalent to $\vec{f}$. $\square$

Remark. The individual components of a Witt vector cannot be truncated to their principal parts one at a time — Witt addition is not componentwise ((2.3)), so a componentwise operation changes the Artin–Schreier–Witt class. The point of Proposition 34 is that all the adjustments are performed by subtracting genuine elements of $\wp W_r(k((x)))$, inside the Witt ring.

The Three Theories Side by Side

The following table summarizes the parallel structure of the three theories over $K = k((x))$, k algebraically closed of characteristic $p > 0$. Each column supplies exactly the ingredients that the corresponding computation in Section 2.5 consumes: the explicit equation (to realize a local torsor over $\mathbb{G}_m$), the reduced representative (to control pole orders), and the ramification criterion (to read off the conclusion at the generic point of the exceptional divisor).

	Kummer	**Artin–Schreier**	**Artin–Schreier–Witt**
group G	$\mathbb{Z}/e\mathbb{Z}$, $\gcd(e,p)=1$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p^r\mathbb{Z}$
ambient group scheme	$\mathbb{G}_m$	$\mathbb{G}_a$	W_r
isogeny	$X \mapsto X^e$	$\wp(X) = X^p - X$	$\wp(\vec{X}) = F\vec{X} \ominus \vec{X}$
defining equation	$X^e = a$	$X^p - X = a$	$\wp(\vec{X}) = \vec{f}$
classifying group	$K^\times/(K^\times)^e$	$K/\wp(K)$	$W_r(K)/\wp W_r(K)$
reduced representative	x^a, $0 \le a < e$	$f \in x^{-1}k[x^{-1}]$, $p \nmid m =$ pole order	$\vec{f} = (f_0, \ldots, f_{r-1})$, $f_j \in x^{-1}k[x^{-1}]$, $p \nmid m_j$
classification	Lemma 15	Lemma 17	Theorem 27, Proposition 34
ramification	tame; ram. index $e/\gcd(a,e)$	totally ramified; unique break m	totally ramified (if $p \nmid m_0$, $f_0 \neq 0$); largest break $\max_j p^{r-1-j} m_j$
criterion	Theorem 14	Theorem 16	Theorem 31
unramified iff	$e \mid a$	$f = 0$	$\vec{f} = \vec{0}$

2.4 Prelude: The Local Geometry of Blowups of Symmetric Powers

This section collects the geometric ingredients that the three computations of Section 2.5 share: the complete local field at the generic point of the exceptional divisor, the formal–local invariance of the whole problem (which reduces us to $\mathbb{P}^1$ and to explicit torsors over $\mathbb{G}_m$), the invariant-theoretic description of the symmetric-power cover, and a regularity criterion for symmetric functions. Each is stated once and reused verbatim in all three cases.

The Blowup of Affine Space at the Origin

Proposition 35. *Let $X = \mathbb{A}^n_k$ be the affine n-space over a field k, let $0 \in X$ be the origin, and let $\tilde{X} = \mathrm{Bl}_0(X)$ be the blowup of X at the origin, so that $\tilde{X} \subset X \times_k \mathbb{P}^{n-1}_k$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}^{n-1}_k$. Let $E \subset \tilde{X}$ be the exceptional divisor — the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, of codimension 1 in $\tilde{X}$ — let $\eta \in E$ be its generic point, and let $R = \mathcal{O}_{\tilde{X},\eta}$. Then:*

1. *R is a discrete valuation ring with fraction field $K = K(X) = k(x_1, \ldots, x_n)$. On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$, the coordinate ring is $\mathcal{O}_{\tilde{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}\right]$, and η corresponds to the prime ideal (x_1), so $R = k[x_1, \frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1}]_{(x_1)}$.*

2. *The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \ldots, \frac{u_n}{u_1})$, and the completion of R is $\hat{R} = \kappa(\eta)[[x_1]]$. The element x_1 is a uniformizer; so is every x_i, since $x_i = (\frac{u_i}{u_1}) x_1$ with $\frac{u_i}{u_1}$ a unit in R.*

3. *The valuation ν_E of R computes the order of vanishing at the origin: for a monomial $M = x_1^{a_1} \ldots x_n^{a_n}$ we have $M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \cdots \left(\frac{u_n}{u_1}\right)^{a_n}$ with the parenthesized factor a unit in R, so*

$\nu_E(M) = \sum a_i = \deg M$; consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$ with f_i homogeneous of degree i and $f_d \neq 0$, we have $\nu_E(f) = d$.

The case relevant to us is the symmetric power: the situation is identical after replacing the coordinates $x_1, \dots, x_d$ by the elementary symmetric polynomials. Let $e_1, \dots, e_d$ be the elementary symmetric polynomials in $x_1, \dots, x_d$, so that $C^{(d)} = \mathrm{Sym}^d_k(\mathbb{A}^1) = \operatorname{Spec} k[e_1, \dots, e_d] \cong \mathbb{A}^d_k$ locally at the origin. Blowing up the origin $Z = V(e_1, \dots, e_d)$ and applying Proposition 35 in the coordinates $e_1, \dots, e_d$ (on the chart $u_d \neq 0$), the local ring at the generic point η of the exceptional divisor is

$$R = k\Big[e_d, \tfrac{u_1}{u_d}, \dots, \tfrac{u_{d-1}}{u_d}\Big]_{(e_d)}, \qquad e_i = \tfrac{u_i}{u_d}\, e_d \ \text{ all uniformizers}, \qquad \tfrac{e_i}{e_d} = \tfrac{u_i}{u_d} \in \kappa(\eta),$$

with residue field $\kappa(\eta) = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d})$, completion $\hat{R} = \kappa(\eta)[[e_d]]$, and complete local field

$$\widehat{K}_\eta = k\Big(\tfrac{e_1}{e_d}, \dots, \tfrac{e_{d-1}}{e_d}\Big)((e_d)). \tag{2.6}$$

Normal Models of Symmetric Powers of Torsors

Let C be a smooth, proper (projective) curve over a field k, and let $U \subset C$ be a dense open subset. Let G be a finite abelian group, and let $\pi_U : U' \to U$ be a G-torsor in $U_{\text{Ét}}$. Let C' be the normal proper model of U' over k (i.e. the unique smooth proper curve which has U' as a dense open subset). Let $\pi : C' \to C$ be the induced morphism (which is finite, generically étale over U). Let $U'^{(d)}$ be the standard d-th symmetric product of U' as a scheme, and let $U'^{[d]}$ be the torsor-theoretic symmetric product; by Corollary 27 we have a canonical morphism $q_U : U'^{(d)} \to U'^{[d]}$. Note that $C'^{(d)}$ is smooth and finite over $C^{(d)}$, hence it is the normalization of $C^{(d)}$ inside $K(C'^{(d)})$. $U'^{(d)}$ is a dense open subset of $C'^{(d)}$, hence $C'^{(d)}$ is a normal proper model of $U'^{(d)}$. Let $C'^{[d]}$ be the normalization of $C^{(d)}$ in the function field $K(U'^{[d]})$; then q_U extends to a map $q : C'^{(d)} \to C'^{[d]}$, and we have the following diagrams:

$$\begin{tikzcd} U' \arrow[r, hook] \arrow[d, "\pi_U"'] & C' \arrow[d, "\pi"] \\ U \arrow[r, hook] & C \end{tikzcd} \qquad \begin{tikzcd} U'^{(d)} \arrow[r, hook] \arrow[d, "q_U"'] & C'^{(d)} \arrow[d, "q"] \\ U'^{[d]} \arrow[r, hook] \arrow[d] & C'^{[d]} \arrow[d] \\ U^{(d)} \arrow[r, hook] & C^{(d)} \end{tikzcd}$$

Lemma 36. *In the context of the preceding discussion, the morphism $U'^{(d)} \to U'^{[d]} \hookrightarrow C'^{[d]}$ is unramified in codimension 1 (see Definition 6).*

Proof. Let $C' \setminus U' = \{x_1, \dots, x_m\}$. For each $x \in C' \setminus U'$, define the locus

$$Z_x = \{D \in C'^{(d)} \mid x \in \mathrm{Supp}(D)\}.$$

Each Z_x is a prime divisor in the symmetric product $C'^{(d)}$, and the boundary $C'^{(d)} \setminus U'^{(d)}$ is precisely the finite union $\bigcup_{i=1}^m Z_{x_i}$.

Let η be the generic point of Z_x. It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]}, q(\eta)} \hookrightarrow \mathcal{O}_{C'^{(d)}, \eta}$ is weakly unramified (Definition 2).

Let $p : C'^d \to C'^{(d)}$ be the natural quotient. Consider the closed subvariety

$$H = \{x\} \times C' \times \cdots \times C' \subset C'^d.$$

The subvariety H is irreducible, and its image under p is exactly Z_x. Consequently, p maps the generic point η_H of H to the generic point η of Z_x.

$$\begin{tikzcd}
\eta_H \arrow[r] \arrow[d] & H \arrow[r] \arrow[d] & C'^d \arrow[d, "p"] \\
\eta \arrow[r] \arrow[d] & Z_x \arrow[r] \arrow[d] & C'^{(d)} \arrow[d, "q"] \\
q(\eta) \arrow[r] & q(Z_x) \arrow[r] & C'^{[d]}
\end{tikzcd}$$

It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]},q(\eta)} \hookrightarrow \mathcal{O}_{C'^d,\eta_H}$ is weakly unramified. Let $R = \mathcal{O}_{C'^d,\eta_H}$, and let $A = \mathcal{O}_{C'^{[d]},q(\eta)}$.

Recall that $C'^{[d]}$ is the quotient of C'^d by the finite group $\Xi = K_G \rtimes S_d$ (see Corollary 27), and the morphism $q \circ p : C'^d \to C'^{[d]}$ is a *branched Galois cover* (finite, surjective map of normal integral schemes $X \to Y$, with function field extension $K(X)/K(Y)$ Galois).

Hence the decomposition group of η_H in Ξ and the inertia group of η_H in Ξ are defined:

$$D_{\eta_H} = \{g \in \Xi \mid g(\eta_H) = \eta_H\}$$

$$I_{\eta_H} = \{g \in D_{\eta_H} \mid g^*(u) \equiv u \pmod{\mathfrak{m}_R} \text{ for all } u \in R\}$$

Claim: The inertia group I_{η_H} is trivial.

Let $K = K(C')$ be the function field of C'; then $K(C'^d) = K^{\otimes_k d} \cong \mathrm{Frac}(K_1 \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ where K_i is the copy of K corresponding to the i-th factor of C'^d. The closed subscheme H corresponds to the prime ideal $\mathfrak{p}_H \subset \mathcal{O}_{C'^d}$, and $R = \mathcal{O}_{C'^d,\mathfrak{p}_H}$. The residue field of R is exactly $K(H) \cong \mathrm{Frac}(k(x) \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ where $k(x)$ is the residue field of x.

The action of $\gamma = (g_1, \ldots, g_d, \sigma) \in \Xi$, via the pullback (γ^*) on $K(C'^d)$ does two things (in that order):

1. It permutes the tensor factors according to σ. A function strictly in K_i is mapped to a function in $K_{\sigma(i)}$.
2. It applies the field automorphism $g_i^* \in \mathrm{Aut}(K/k)$ to the respective factors.

Let $\gamma \in I_{\eta_H} \subset \Xi$. Then the induced automorphism $\bar{\gamma}^*$ on $K(H)$ must be the identity map. This implies that σ must fix i for every $i \geq 2$ (the K_i are linearly independent), and hence $\sigma = id$. The trivial action of $\bar{\gamma}^*$ restricts to a trivial action of g_i^* on K_i for every $i \geq 2$, and hence $g_i = 1$ for every $i \geq 2$ (since the action of G on $K(C')$ is faithful). Finally, since $g_1 g_2 \cdots g_d = 1$, we also have $g_1 = 1$. Since the inertia group I_{η_H} is trivial, the local ring extension $A \hookrightarrow R$ is weakly unramified. (See for example [Stacks, Tag 09E3], [Stacks, Tag 0BTF])

□

Formal–Local Invariance and Reduction to $\mathbb{P}^1$

The next two results make precise the statement that *the ramification of* $\mathcal{P}^{[d]}$ *at* $\eta_{\mathfrak{n}}$ *is a formal–local invariant of* $\mathcal{P}$ *at* P: it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\operatorname{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion; for blowups, this is Theorem 30. We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately.

Throughout the rest of this section we assume that k is algebraically closed; in particular every closed point of C is k-rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark 37. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ along $Z_{\mathfrak{n}}$, and of $\mathcal{P}^{[d]}$ (Corollary 27) commutes with the flat base change $\operatorname{Spec}(\bar{k}) \to \operatorname{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}_k^{\deg \mathfrak{n}-1}$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}},\bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. The proof of Lemma 11 uses only these two properties of the extension (there the extension was moreover finite étale), so it applies verbatim: the ramification of $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ at $\bar{\eta}$ is bounded by r if and only if the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is. Hence unramifiedness and tameness at $\eta_{\mathfrak{n}}$ may be checked after base change to $\bar{k}$.

2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $Z_{\mathfrak{n}}$ corresponds to the divisor $\sum_{\bar{P}\mapsto P} n\bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P) : k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 1 under the standing assumption that k is algebraically closed; this suffices for our purposes, since in the proof of Theorem 76 the field is assumed algebraically closed before the present theorem is invoked.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C,P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. The scheme $\operatorname{Spec}(k[[x]])$ has two points — the closed point, mapping to P, and the generic point, mapping to the generic point of C — so the punctured formal disc $\operatorname{Spec}(k((x)))$ maps to U, and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \operatorname{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G-torsor over $\operatorname{Spec}(k((x)))$. Let $e_1, \dots, e_d \in k[[x_1, \dots, x_d]]$ be the elementary symmetric polynomials in $x_1, \dots, x_d$, and set

$$V = \operatorname{Spec}\Big(k[[e_1, \dots, e_d]][e_d^{-1}]\Big), \qquad W = \operatorname{Spec}\Big(k[[x_1, \dots, x_d]][(x_1 \cdots x_d)^{-1}]\Big).$$

The inclusion $k[[e_1, \dots, e_d]] \subset k[[x_1, \dots, x_d]]$ induces a finite flat morphism $\hat{r} : W \to V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \to \operatorname{Spec}(k((x))), \qquad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \dots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$.

Lemma 38 (Formal–local invariance). *In the above notation:*

1. *The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)},Z_\mathfrak{n}}} \cong k[[e_1,\dots,e_d]]$ carrying the maximal ideal to $(e_1,\dots,e_d)$, such that the induced morphism $c : \operatorname{Spec}(k[[e_1,\dots,e_d]]) \to C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \to U^{(d)}$ for the restriction.*

2. *There is an induced isomorphism $\widehat{\mathcal{O}_{X_\mathfrak{n},\eta_\mathfrak{n}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d},\dots,\frac{e_{d-1}}{e_d})$. In particular $\widehat{K}_{\eta_\mathfrak{n}} := \operatorname{Frac}(\widehat{\mathcal{O}_{X_\mathfrak{n},\eta_\mathfrak{n}}}) \cong \kappa((e_d))$, and the canonical morphism $\operatorname{Spec}(\widehat{K}_{\eta_\mathfrak{n}}) \to U^{(d)}$ factors as $\operatorname{Spec}(\kappa((e_d))) \to V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C,\mathcal{P})$.*

3. *There is a canonical isomorphism of G-torsors over V*

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \;\cong\; \Xi\backslash\Big(q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x\Big).$$

Consequently, the G-torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec}(\widehat{K}_{\eta_\mathfrak{n}})}$ — and hence whether the ramification of $\mathcal{P}^{[d]}$ at $\eta_\mathfrak{n}$ is bounded by r (resp. tame, resp. unramified) — depends only on d and on the isomorphism class of the G-torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C,\mathcal{P})$.

Proof. **(1)** Since P is a k-rational point of the smooth curve C, the uniformizer x induces $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. As completion at rational points takes products of k-varieties to completed tensor products, we get $\widehat{\mathcal{O}_{C^d,(P,\dots,P)}} \cong k[[x]]\widehat{\otimes}_k \cdots \widehat{\otimes}_k k[[x]] = k[[x_1,\dots,x_d]]$. Write $A = \mathcal{O}_{C^{(d)},Z_\mathfrak{n}}$ and $B = \mathcal{O}_{C^d,(P,\dots,P)}$. Since k is algebraically closed, a closed point $(Q_1,\dots,Q_d) \in C^d$ maps to $Z_\mathfrak{n} = dP$ under the finite projection $r : C^d \to C^{(d)}$ if and only if $Q_1 = \cdots = Q_d = P$; hence $(P,\dots,P)$ is the unique point of C^d over $Z_\mathfrak{n}$, $B = (r_*\mathcal{O}_{C^d})_{Z_\mathfrak{n}}$ is a finite A-module, and $A = B^{S_d}$ (formation of invariants commutes with localization at invariant primes). Since $\mathfrak{m}_B$ is the unique maximal ideal of B over $\mathfrak{m}_A$, we have $\mathfrak{m}_B^N \subset \mathfrak{m}_A B \subset \mathfrak{m}_B$ for some N, so the $\mathfrak{m}_A$-adic and $\mathfrak{m}_B$-adic topologies on B agree, and therefore $B \otimes_A \hat{A} \cong \hat{B} \cong k[[x_1,\dots,x_d]]$. Now $A = B^{S_d}$ is the kernel of $B \to \prod_{\sigma\in S_d} B$, $b \mapsto (\sigma(b) - b)_\sigma$; as $\hat{A}$ is flat over A, tensoring with $\hat{A}$ preserves this kernel, so

$$\hat{A} \cong (B \otimes_A \hat{A})^{S_d} \cong k[[x_1,\dots,x_d]]^{S_d}.$$

Finally, $k[[x_1,\dots,x_d]]^{S_d} = k[[e_1,\dots,e_d]]$: the ring $k[x_1,\dots,x_d]$ is a finite free module over $k[e_1,\dots,e_d]$, the same topological argument identifies $k[[x_1,\dots,x_d]]$ with $k[x_1,\dots,x_d] \otimes_{k[e_1,\dots,e_d]} k[[e_1,\dots,e_d]]$, and taking S_d-invariants — again a kernel, preserved by the flat base change $k[e_1,\dots,e_d] \to k[[e_1,\dots,e_d]]$ — yields $k[[e_1,\dots,e_d]]$. Under these identifications the maximal ideal of $\hat{A}$ is $(e_1,\dots,e_d)$, and c is flat as the composition of the flat morphisms $\operatorname{Spec}(\hat{A}) \to \operatorname{Spec}(A) \to C^{(d)}$.

It remains to check that $c^{-1}(C^{(d)} \setminus U^{(d)}) = V(e_d)$ as sets. We have $C^{(d)} \setminus U^{(d)} = \bigcup_{Q\in\operatorname{Supp}(\mathfrak{m})} Z_Q$ with $Z_Q = \{D : Q \in \operatorname{Supp}(D)\}$, as in the proof of Lemma 36. For $Q \neq P$ the divisor Z_Q does not contain the point $Z_\mathfrak{n} = dP$, so its ideal is not contained in $\mathfrak{m}_A$, and $c^{-1}(Z_Q) = \emptyset$. For $Q = P$, pull back along the finite surjective morphism $\operatorname{Spec}(\hat{B}) \to \operatorname{Spec}(\hat{A})$: the preimage of Z_P in C^d is $\bigcup_i p_i^{-1}(P)$, whose trace on $\operatorname{Spec}(\hat{B}) = \operatorname{Spec}(k[[x_1,\dots,x_d]])$ is $V(x_1) \cup \cdots \cup V(x_d) = V(x_1\cdots x_d) = V(e_d)$; by surjectivity, $c^{-1}(Z_P) = V(e_d)$ as claimed. Hence $c^{-1}(U^{(d)}) = V$, and likewise $\operatorname{Spec}(\hat{B}) \times_{C^d} U^d = W$.

(2) Blowing up commutes with flat base change, and the center $Z_\mathfrak{n}$ pulls back under c to the closed point $V(e_1,\dots,e_d)$; hence

$$\hat{Y} := \operatorname{Bl}_{(e_1,\dots,e_d)}\big(\operatorname{Spec} k[[e_1,\dots,e_d]]\big) \cong X_\mathfrak{n} \times_{C^{(d)}} \operatorname{Spec}(\hat{A}),$$

and the exceptional divisor $\hat{E} \subset \hat{Y}$ is the base change of $E_{\mathfrak{n}}$ along $\mathrm{Spec}(\hat{A}/\hat{\mathfrak{m}}) = \mathrm{Spec}(\kappa(Z_{\mathfrak{n}}))$, i.e. the projection $\mathrm{pr} : \hat{Y} \to X_{\mathfrak{n}}$ restricts to an isomorphism $\hat{E} \cong E_{\mathfrak{n}} \cong \mathbb{P}^{d-1}_k$. Let $\hat{\eta}$ be the generic point of $\hat{E}$. Then $\mathrm{pr}(\hat{\eta}) = \eta_{\mathfrak{n}}$, and the induced local homomorphism of discrete valuation rings $\mathcal{O}_{X_{\mathfrak{n}},\eta_{\mathfrak{n}}} \to \mathcal{O}_{\hat{Y},\hat{\eta}}$ carries a uniformizer to a uniformizer (the ideal of $E_{\mathfrak{n}}$ pulls back to the ideal of $\hat{E}$) and induces an isomorphism of residue fields (both are the function field of $\mathbb{P}^{d-1}_k$); hence it induces an isomorphism on completions. The completion of $\mathcal{O}_{\hat{Y},\hat{\eta}}$ is computed exactly as in Proposition 35 (in the coordinates $e_1, \dots, e_d$, on the chart $u_d \neq 0$): we have $e_i = \frac{u_i}{u_d} e_d$, the local ring at $\hat{\eta}$ is $\Big(\hat{A}[\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d}]\Big)_{(e_d)}$, its residue field is $\kappa = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d}) = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d})$, and its completion is $\kappa[[e_d]]$. Finally, e_d is invertible in $\widehat{K}_{\eta_{\mathfrak{n}}} = \kappa((e_d))$, so the composite $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to \mathrm{Spec}(\mathcal{O}_{\hat{Y},\hat{\eta}}) \to \hat{Y} \to \mathrm{Spec}(\hat{A})$ factors through V, giving the asserted factorization of $\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}}) \to U^{(d)}$ through c_V.

(3) Let $h : \mathcal{P}^{\times_k d} \to U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \to U^d$ with the finite morphism $U^d \to U^{(d)}$. By Corollary 27, $\mathcal{P}^{[d]} = \Xi \backslash \mathcal{P}^{\times_k d} = \underline{\mathrm{Spec}}_{U^{(d)}}\Big((h_* \mathcal{O}_{\mathcal{P}^{\times_k d}})^{\Xi}\Big)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \to U^{(d)}$, and the formation of Ξ-invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi \backslash \Big(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V\Big).$$

By step (1), $U^d \times_{U^{(d)}} V \cong W$, so $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$. For each i, the composition $W \to \mathrm{Spec}(k[[x_1, \dots, x_d]]) \to \mathrm{Spec}(k[[x]]) \to C$ — the middle map being the completion of the projection $p_i : C^d \to C$, i.e. $x \mapsto x_i$ — factors through $\mathrm{Spec}(k((x))) \to U$, since x_i is invertible on W; this composite is precisely q_i, and it commutes with $p_i : U^d \to U$. Since $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$, base changing to W gives an isomorphism

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i, compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ-quotients yields the displayed isomorphism.

For the final statement: the schemes V and W, the maps q_i and $\hat{r}$, the Ξ-action, and the point $\mathrm{Spec}(\kappa((e_d))) \to V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ is the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since the ramification at $\eta_{\mathfrak{n}}$ is by definition read off from $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{\mathfrak{n}}})}$ together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 6, Definition 7), it depends only on $(\mathcal{P}_x, d)$. □

Corollary 39 (Reduction to $\mathbb{P}^1$). *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k, let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \to U' = C' \setminus \mathfrak{m}'$ be G-torsors, let $P \in \mathrm{Supp}(\mathfrak{m})(k)$ and $P' \in \mathrm{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G-torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then for every $d \geq 1$ and every r, the ramification of $\mathcal{P}^{[d]}$ at η_{dP} is bounded by r if and only if the ramification of $\mathcal{P}'^{[d]}$ at $\eta_{dP'}$ is; in particular, the one is unramified (resp. tamely ramified) at η_{dP} if and only if the other is at $\eta_{dP'}$.*

Proof. Immediate from Lemma 38: both $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\mathrm{Spec}(\kappa((e_d))) \to V$ of $\Xi \backslash (q_1^{-1}\mathcal{P}_x \times_W \cdots \times_W q_d^{-1}\mathcal{P}_x)$. □

The Invariant-Theory Scaffold

All three computations of Section 2.5 identify the symmetric-power cover at $\eta_{\mathfrak{n}}$ by the same invariant-theoretic mechanism, which we set up once. Let G be a finite abelian group and let $\mathcal{P} = \operatorname{Spec} S \to \mathbb{G}_m = \operatorname{Spec} R$, $R = k[x, x^{-1}]$, be an affine G-torsor. Write

$$R^{\otimes_k d} = k[x_1^{\pm 1}, \ldots, x_d^{\pm 1}],$$

the coordinate ring of $U^d = \mathbb{G}_m^d$, and let $S^{\otimes_k d}$ be the coordinate ring of the G^d-torsor $\mathcal{P}^{\times_k d} = p_1^{-1}\mathcal{P} \times_{U^d} \cdots \times_{U^d} p_d^{-1}\mathcal{P}$ over U^d.

The d-fold product torsor $p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P}$ on U^d — the contracted product along the sum map $G^d \to G$ — is the quotient of $\mathcal{P}^{\times_k d}$ by the kernel $K_G = \{(g_1, \ldots, g_d) \in G^d \mid g_1 + \cdots + g_d = 0\}$ of the sum map. We identify $K_G \cong G^{d-1}$ via $(g_1, \ldots, g_{d-1}) \mapsto (g_1,\, g_2 - g_1,\, \ldots,\, -g_{d-1})$; its coordinate ring is the invariant ring $\left(S^{\otimes_k d}\right)^{G^{d-1}}$, and the residual action of $G = G^d/K_G$ is induced by $(g, 0, \ldots, 0) \in G^d$. The symmetric group S_d acts on everything by permuting the d factors, and by Corollary 27 the symmetric-power torsor is the $\Xi = K_G \rtimes S_d$ quotient:

$$\mathcal{P}^{[d]}\big|_{U^{(d)}} = \operatorname{Spec}\left(\left(S^{\otimes_k d}\right)^{\Xi}\right) = \operatorname{Spec}\left(\left(\left(S^{\otimes_k d}\right)^{G^{d-1}}\right)^{S_d}\right).$$

On the base, the S_d-invariants form the ring of symmetric Laurent polynomials

$$\left(R^{\otimes_k d}\right)^{S_d} = k[e_1, e_2, \ldots, e_d, e_d^{-1}],$$

where $e_1, \ldots, e_d$ are the elementary symmetric polynomials in $x_1, \ldots, x_d$ (a Laurent polynomial is symmetric if and only if, after clearing the denominator $e_d^N = (x_1 \cdots x_d)^N$, its numerator is a symmetric polynomial). Finally, by Lemma 38(2)–(3) applied to $C = \mathbb{P}^1$, $\mathfrak{m} = d \cdot 0 + \infty$, the restriction of $\mathcal{P}^{[d]}$ to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$ of (2.6) is the base change of $\operatorname{Spec}\left(\left(S^{\otimes_k d}\right)^{\Xi}\right)$ along $k[e_1, \ldots, e_d, e_d^{-1}] \to \widehat{K}_\eta$. In each of the three computations, the task is therefore to identify the invariant ring $\left(S^{\otimes_k d}\right)^{\Xi}$ explicitly as the algebra of an equation of the same shape as that of $\mathcal{P}$, with datum a *symmetric* function of the x_i, and then to bound the valuation of that datum in $\widehat{K}_\eta$.

Compatibilities of the Symmetric Power

Recall from Corollary 27 that $\mathcal{P}^{[d]} = \Xi_G \backslash \mathcal{P}^{\times_k d}$, where $\Xi_G = K_G \rtimes S_d$. The symmetric power is compatible with the two standard operations on the structure group: fiber products and induction along injections.

Lemma 40 (symmetric powers and fiber products)**.** *Let G_1, G_2 be finite abelian groups, let $\mathcal{P}_1, \mathcal{P}_2$ be a G_1-, resp. G_2-torsor on $U_{\acute{e}t}$, and let $\mathcal{P} = \mathcal{P}_1 \times_U \mathcal{P}_2$ be the associated $(G_1 \times G_2)$-torsor. Then there is a canonical isomorphism of $(G_1 \times G_2)$-torsors over $U^{(d)}$*

$$\mathcal{P}^{[d]} \;\cong\; \mathcal{P}_1^{[d]} \times_{U^{(d)}} \mathcal{P}_2^{[d]}.$$

Proof. Write $G = G_1 \times G_2$. We have $\mathcal{P}^{\times_k d} = \mathcal{P}_1^{\times_k d} \times_{U^d} \mathcal{P}_2^{\times_k d}$, and under this identification $K_G = K_{G_1} \times K_{G_2}$ inside $G^d = G_1^d \times G_2^d$, with S_d acting diagonally. Consider the composition

$$\mathcal{P}^{\times_k d} \longrightarrow \mathcal{P}_1^{\times_k d} \times_{U^d} \mathcal{P}_2^{\times_k d} \longrightarrow \left(\Xi_{G_1} \backslash \mathcal{P}_1^{\times_k d}\right) \times_{U^{(d)}} \left(\Xi_{G_2} \backslash \mathcal{P}_2^{\times_k d}\right) = \mathcal{P}_1^{[d]} \times_{U^{(d)}} \mathcal{P}_2^{[d]}.$$

The first projection $\mathcal{P}^{\times_k d} \to \mathcal{P}_1^{[d]}$ is invariant under $K_{G_1} \rtimes S_d$ by construction and under K_{G_2} because K_{G_2} acts only on the second factor; symmetrically for the second projection. Hence the composition is Ξ_G-invariant and descends to a morphism

$$\phi\colon \mathcal{P}^{[d]} = \Xi_G \backslash \mathcal{P}^{\times_k d} \longrightarrow \mathcal{P}_1^{[d]} \times_{U^{(d)}} \mathcal{P}_2^{[d]}$$

over $U^{(d)}$. The residual G-action on the source is induced by $(g_1, g_2, 0, \ldots, 0) \in G^d$, which maps under the projections to the residual G_1- and G_2-actions on the factors of the target; so ϕ is G-equivariant. The source is a G-torsor (Corollary 27), and the target is a fiber product of a G_1-torsor and a G_2-torsor, hence a G-torsor. A G-equivariant morphism of G-torsors is an isomorphism (Proposition 5). □

Lemma 41 (symmetric powers and induced torsors)**.** *Let $\iota : H \hookrightarrow G$ be an injective homomorphism of finite abelian groups and let $\mathcal{Q}$ be an H-torsor on $U_{\acute{e}t}$. Then there is a canonical isomorphism of G-torsors over $U^{(d)}$*

$$(\iota_* \mathcal{Q})^{[d]} \;\cong\; \iota_*\big(\mathcal{Q}^{[d]}\big).$$

Proof. The canonical map $j\colon \mathcal{Q} \to \iota_*\mathcal{Q} = (\mathcal{Q} \times G)/H$, $q \mapsto [(q, 1)]$, satisfies $j(qh) = j(q) \cdot \iota(h)$. Its d-fold product $\mathcal{Q}^{\times_k d} \to (\iota_*\mathcal{Q})^{\times_k d}$ is equivariant for $H^d \to G^d$ (which carries K_H into K_G) and commutes with the S_d-actions, hence descends to an H-equivariant morphism of the quotients

$$\psi\colon \mathcal{Q}^{[d]} = \Xi_H \backslash \mathcal{Q}^{\times_k d} \longrightarrow \Xi_G \backslash (\iota_*\mathcal{Q})^{\times_k d} = (\iota_*\mathcal{Q})^{[d]},$$

where the residual $H = H^d/K_H$ acts on the target through $\iota : H \to G = G^d/K_G$. By the universal property of induction, an H-equivariant morphism from the H-torsor $\mathcal{Q}^{[d]}$ to (the restriction of) the G-torsor $(\iota_*\mathcal{Q})^{[d]}$ extends uniquely to a G-equivariant morphism

$$\iota_*\big(\mathcal{Q}^{[d]}\big) \longrightarrow (\iota_*\mathcal{Q})^{[d]}, \qquad [(z, g)] \longmapsto \psi(z) \cdot g,$$

which is well defined precisely because ψ is H-equivariant. A G-equivariant morphism of G-torsors over $U^{(d)}$ is an isomorphism (Proposition 5). □

A Symmetric Regularity Lemma

The valuation bound is in all cases an instance of the following lemma, applied with $y_i := 1/x_i$. Note that the elementary symmetric polynomials in the y_i are related to those in the x_i by

$$e_k(y_1, \ldots, y_d) = \sum_{1 \le i_1 < \cdots < i_k \le d} \frac{1}{x_{i_1} \cdots x_{i_k}} = \frac{e_{d-k}(x_1, \ldots, x_d)}{e_d(x_1, \ldots, x_d)}, \tag{2.7}$$

which for $1 \le k \le d-1$ lies in the residue field $\kappa(\eta) = k(e_1/e_d, \ldots, e_{d-1}/e_d)$ of (2.6).

Lemma 42 (symmetric regularity)**.** *Let $\Phi \in k[y_1, \ldots, y_d]^{S_d}$ be symmetric with every monomial of total degree $< d$. Then $\Phi \in \kappa(\eta) = k(e_1/e_d, \ldots, e_{d-1}/e_d)$; in particular Φ is regular at $\eta_{\mathfrak{n}}$.*

Proof. Φ is a k-combination of monomial symmetric functions $m_\lambda(y)$ with $|\lambda| < d$. Each $m_\lambda(y)$ is an isobaric polynomial in the elementary symmetric functions $e_\mu(y) = \prod_i e_{\mu_i}(y)$ with $\mu \vdash |\lambda|$; every part $\mu_i \le |\lambda| < d$, so only $e_1(y), \ldots, e_{d-1}(y)$ occur — never $e_d(y)$. By (2.7), each $e_k(y)$ with $1 \le k \le d-1$ lies in $\kappa(\eta)$. Hence $\Phi \in \kappa(\eta) \subset \widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$. □

2.5 The Symmetric-Power Computations

We now prove Theorem 1. The proof proceeds through the three cyclic cases — $G = \mathbb{Z}/e\mathbb{Z}$ with $\gcd(e,p) = 1$, $G = \mathbb{Z}/p\mathbb{Z}$, and $G = \mathbb{Z}/p^r\mathbb{Z}$ — each treated by the *same* three-step computation, one column of the table of Section 2.3 at a time:

1. **Realize.** The restriction $\mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc at P is realized, using the reduced representatives of Section 2.3, by an explicit equation over $\mathbb{G}_m \subset \mathbb{P}^1$; by Corollary 39 we may then assume $C = \mathbb{P}^1_k$, $\mathfrak{n} = d \cdot 0$, $U = \mathbb{G}_m$, and $\mathcal{P}$ is this explicit torsor.
2. **Identify.** Using the invariant-theory scaffold of Section 2.4, the restriction of $\mathcal{P}^{[d]}$ to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$ of (2.6) is identified as the cover defined by an equation *of the same shape*, whose datum is the symmetrization of the original datum: the product $\prod_i f(x_i)$ in the Kummer case, the sum $\sum_i f(x_i)$ in the Artin–Schreier case, and the Witt sum $\bigoplus_i \vec{f}(x_i)$ in the Artin–Schreier–Witt case.
3. **Count.** The valuation of the symmetrized datum in $\widehat{K}_\eta$ is bounded — using Lemma 42 and the pole bounds built into the reduced representative — and the ramification criterion of the relevant theory yields the conclusion: *tame* in the Kummer case, *unramified* in the two wild cases.

Finally, in Section 2.5.4, the general finite abelian case is assembled from the three cyclic cases by decomposing G into primary cyclic factors, using the compatibility lemmas of Section 2.4.

Throughout this section k is algebraically closed (Remark 37), $P \in C(k)$, $\mathfrak{n} = nP$, and $d = \deg \mathfrak{n} = n$.

2.5.1 The Tame Case

Step 1: realization over $\mathbb{G}_m$

Lemma 43 (Realization over $\mathbb{G}_m$, Kummer case). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a tamely ramified G-torsor over* $\operatorname{Spec}(k((x)))$*. Then there exist an integer e prime to p, an injective homomorphism $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$, and an integer a prime to e, such that*

$$\mathcal{P}'_0 := \operatorname{Spec}\Big(k[x, x^{-1}][T]/(T^e - x^a)\Big)$$

is a $\mathbb{Z}/e\mathbb{Z}$-torsor over $\mathbb{G}_m$, tamely ramified at 0 and ∞, and the induced G-torsor $\mathcal{P}' := \iota_* \mathcal{P}'_0$ satisfies $\mathcal{P}'_x \cong \mathcal{Q}$. Here $T^e = x^a$ is regarded as a $\mathbb{Z}/e\mathbb{Z}$-torsor via a fixed primitive e-th root of unity $\zeta_e \in k$.

Proof. By Lemma 15(1), the tame quotient G^t of $\operatorname{Gal}(k((x))^{sep}/k((x)))$ is procyclic; fix a topological generator γ. Let $\rho : \operatorname{Gal}(k((x))^{sep}/k((x))) \to G$ be the continuous homomorphism classifying $\mathcal{Q}$. Tameness means that ρ kills the wild inertia subgroup, so ρ factors through G^t and is determined by $g := \rho(\gamma)$, whose order e is prime to p. Let $\iota : \mathbb{Z}/e\mathbb{Z} \xrightarrow{\sim} \langle g \rangle \subset G$ be given by $1 \mapsto g$, so that $\rho = \iota \circ \chi$ for a surjective character $\chi : G^t \to \mathbb{Z}/e\mathbb{Z}$. By Lemma 15(2), the classes of x^a, $a \in \mathbb{Z}/e\mathbb{Z}$, exhaust $\operatorname{Hom}_{\text{cont.}}(G^t, \mathbb{Z}/e\mathbb{Z}) \cong \mathbb{Z}/e\mathbb{Z}$, the class of x^a being surjective precisely when $\gcd(a,e) = 1$. Choose a with the class of x^a equal to χ; then $\mathcal{Q} \cong \iota_*(T^e = x^a)$ over $k((x))$. The same equation over $k[x, x^{-1}]$ defines a $\mathbb{Z}/e\mathbb{Z}$-torsor $\mathcal{P}'_0$ over $\mathbb{G}_m$ (étale since e is invertible in k), with $(\iota_* \mathcal{P}'_0)_x \cong \mathcal{Q}$; it is tamely ramified at 0 and at ∞ since its local characters there factor through the tame quotients. $\square$

Step 2: identification of the symmetric-power cover

Proposition 44 (identification of the product torsor, Kummer case)**.** *Let $G = \mathbb{Z}/e\mathbb{Z}$ with e prime to p, let $f \in R^\times$ be a unit of $R = k[x, x^{-1}]$, and let $\mathcal{P}' = \operatorname{Spec} S \to \mathbb{G}_m$, $S = R[X]/(X^e - f)$, be the associated G-torsor. In the notation of the scaffold of Section 2.4, set*

$$Y := X_1 X_2 \cdots X_d \in S^{\otimes_k d}, \qquad F := \prod_{i=1}^{d} f(x_i) \in R^{\otimes_k d}.$$

Then:

1. *Y is K_G-invariant, and*
$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \;\cong\; R^{\otimes_k d}[Y]/\left(Y^e - F\right)$$
as G-torsors over $R^{\otimes_k d}$ (the residual G acting by $Y \mapsto \zeta_e^g Y$).

2. *F is a symmetric Laurent polynomial, i.e. lies in $\left(R^{\otimes_k d}\right)^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}]$, and*
$$\left(\left(S^{\otimes_k d}\right)^{G^{d-1}}\right)^{S_d} \;\cong\; k[e_1, \ldots, e_d, e_d^{-1}][Y]/\left(Y^e - F\right).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$ of (2.6),

$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \;\cong\; \operatorname{Spec}\; \widehat{K}_\eta[Z]/\left(Z^e - F\right).$$

Proof. 1 Recall that $g \in G = \mathbb{Z}/e\mathbb{Z}$ acts on X by $g(X) = \zeta_e^g X$. Hence $(g_1, \ldots, g_{d-1}) \in K_G \cong G^{d-1}$ acts on the generators by

$$X_1 \mapsto \zeta_e^{g_1} X_1, \qquad X_i \mapsto \zeta_e^{g_i - g_{i-1}} X_i \;(1 < i < d), \qquad X_d \mapsto \zeta_e^{-g_{d-1}} X_d,$$

so $Y = X_1 \cdots X_d$ transforms by the telescoping product $\zeta_e^{g_1 + (g_2 - g_1) + \cdots - g_{d-1}} = 1$: it is invariant. Moreover $Y^e = \prod_i X_i^e = \prod_i f(x_i) = F$. Since F is a unit of $R^{\otimes_k d}$ and e is invertible in k, $T := R^{\otimes_k d}[Y]/(Y^e - F)$ is a $\mathbb{Z}/e\mathbb{Z}$-torsor over U^d (finite étale, with G acting by $Y \mapsto \zeta_e^g Y$), and $Y \mapsto X_1 \cdots X_d$ defines an $R^{\otimes_k d}$-algebra homomorphism $\phi : T \to \left(S^{\otimes_k d}\right)^{G^{d-1}}$. The target is the d-fold product torsor (the contracted product of the $p_i^{-1}\mathcal{P}'$ along the sum map $G^d \to G$), a G-torsor over U^d by Proposition 25 and the surrounding discussion; its residual G-action is induced by $(g, 0, \ldots, 0) \in G^d$, under which $Y \mapsto \zeta_e^g Y$. So $\operatorname{Spec} \phi$ is a G-equivariant morphism of G-torsors over U^d, hence an isomorphism by Proposition 5.

2 S_d permutes the x_i and the X_i, so it fixes Y and F; in particular $F \in \left(R^{\otimes_k d}\right)^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}]$ (as recorded in the scaffold). T is free over $R^{\otimes_k d}$ with basis the S_d-invariant monomials Y^j, $0 \le j < e$; writing an element in this basis, it is S_d-invariant if and only if all its coefficients are, whence

$$T^{S_d} = \left(R^{\otimes_k d}\right)^{S_d}[Y]/\left(Y^e - F\right) = k[e_1, \ldots, e_d, e_d^{-1}][Y]/\left(Y^e - F\right).$$

The final statement follows from the scaffold: by Corollary 27, $\mathcal{P}'^{[d]}$ is the Ξ-quotient computed above, and its restriction to $\operatorname{Spec} \widehat{K}_\eta$ along $k[e_1, \ldots, e_d, e_d^{-1}] \to \widehat{K}_\eta$ (Lemma 38) is the Kummer cover of the image of F. $\square$

Step 3: the valuation count and the conclusion

Theorem 45 (the tame case)**.** *Let G be a finite abelian group and let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ be a G-torsor which is tamely ramified at P. Then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

Proof. Realize. The restriction $\mathcal{Q} = \mathcal{P}_x$ is tamely ramified, so Lemma 43 produces e prime to p, $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$, a prime to e, and the torsor $\mathcal{P}'_0 : T^e = x^a$ over $\mathbb{G}_m$ with $(\iota_*\mathcal{P}'_0)_x \cong \mathcal{P}_x$. By Corollary 39, the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ agrees with that of $(\iota_*\mathcal{P}'_0)^{[d]}$ at $\eta_{d\cdot 0}$. By Lemma 41, $(\iota_*\mathcal{P}'_0)^{[d]} \cong \iota_*\big(\mathcal{P}'^{[d]}_0\big)$, and by Lemma 13 it is tamely ramified at $\eta_{d\cdot 0}$ if and only if $\mathcal{P}'^{[d]}_0$ is. So we may assume $G = \mathbb{Z}/e\mathbb{Z}$ and $\mathcal{P} = \mathcal{P}'_0$.

Identify. Proposition 44, with $f = x^a$, identifies

$$\mathcal{P}'^{[d]}_0\big|_{\operatorname{Spec}\widehat{K}_\eta} \cong \operatorname{Spec}\, \widehat{K}_\eta[Z]/\big(Z^e - F\big), \qquad F = \prod_{i=1}^{d} x_i^a = (x_1 \cdots x_d)^a = e_d^a.$$

Count. In $\widehat{K}_\eta = \kappa(\eta)((e_d))$ we have $v_{\widehat{K}_\eta}(F) = v_{\widehat{K}_\eta}(e_d^a) = a$. By Theorem 14 (all of whose cases are tamely ramified, since $\gcd(e,p) = 1$), the extension defined by $Z^e = e_d^a$ is tamely ramified — explicitly, its ramification index is $e/\gcd(a,e)$, prime to p. Hence $\mathcal{P}'^{[d]}_0$, and therefore $\mathcal{P}^{[d]}$, is tamely ramified at $\eta_{\mathfrak{n}}$. □

2.5.2 The Wild Case $G = \mathbb{Z}/p\mathbb{Z}$

Step 1: realization over $\mathbb{G}_m$

Lemma 46 (Realization over $\mathbb{G}_m$, Artin–Schreier case)**.** *Let k be algebraically closed of characteristic $p > 0$, let $G = \mathbb{Z}/p\mathbb{Z}$, and let $\mathcal{Q}$ be a G-torsor over* $\operatorname{Spec}(k((x)))$ *with ramification bounded by d (Definition 7). Then there exists $f \in x^{-1}k[x^{-1}]$, either $f = 0$ or $f = cx^{-m} + a_{-m+1}x^{-m+1} + \cdots + a_{-1}x^{-1}$ with $c \neq 0$, $p \nmid m$ and $m < d$, such that*

$$\mathcal{P}' := \operatorname{Spec}\Big(k[x, x^{-1}][X]/(X^p - X - f)\Big)$$

is a $\mathbb{Z}/p\mathbb{Z}$-torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$. Moreover $\mathcal{P}'$ extends to a torsor over $\mathbb{P}^1_k \setminus \{0\}$ — in particular it is unramified at ∞ — and it has ramification bounded by d at 0.

Proof. If $\mathcal{Q}$ is trivial, take $f = 0$. Otherwise, since $\mathbb{Z}/p\mathbb{Z}$ is simple and $k((x))$ admits no non-trivial unramified extensions, $\mathcal{Q} = \operatorname{Spec}(F)$ for a cyclic extension $F/k((x))$ of degree p together with an isomorphism $\operatorname{Gal}(F/k((x))) \cong \mathbb{Z}/p\mathbb{Z}$. By Artin–Schreier theory (Theorem 16) such data correspond to non-zero classes in $k((x))/\wp(k((x)))$, and by Lemma 17 the class of $\mathcal{Q}$ admits a reduced representative $f \in x^{-1}k[x^{-1}]$, non-zero with pole order m prime to p. The unique ramification break of $F = k((x))[X]/(X^p - X - f)$ is at m (Theorem 16(3)), so the ramification of $\mathcal{Q}$ is bounded by d if and only if $m < d$. Finally, $f \in k[x, x^{-1}]$, so the same equation defines a finite étale $\mathbb{Z}/p\mathbb{Z}$-cover $\mathcal{P}' \to \mathbb{G}_m$ (a torsor via $X \mapsto X + 1$) with $\mathcal{P}'_x \cong \mathcal{Q}$; and writing $y = x^{-1}$, we have $f \in y\,k[y]$, so $X^p - X - f$ defines a finite étale cover of $\mathbb{A}^1_k = \operatorname{Spec}(k[y]) = \mathbb{P}^1_k \setminus \{0\}$ extending $\mathcal{P}'$, whence $\mathcal{P}'$ is unramified at ∞. □

Step 2: identification of the symmetric-power cover

Proposition 47 (identification of the product torsor, Artin–Schreier case)**.** *Let $G = \mathbb{Z}/p\mathbb{Z}$, let $f \in R = k[x, x^{-1}]$, and let $\mathcal{P}' = \operatorname{Spec} S \to \mathbb{G}_m$, $S = R[X]/(X^p - X - f)$, be the associated G-torsor. In the notation of the scaffold, set*

$$Y := X_1 + X_2 + \cdots + X_d \in S^{\otimes_k d}, \qquad \alpha := \sum_{i=1}^{d} f(x_i) \in R^{\otimes_k d}.$$

Then:

1. *Y is K_G-invariant, and*

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \;\cong\; R^{\otimes_k d}[Y]/\left(Y^p - Y - \alpha\right)$$

   as G-torsors over $R^{\otimes_k d}$ (the residual G acting by $Y \mapsto Y + g$).

2. *α is a symmetric Laurent polynomial, i.e. lies in $\left(R^{\otimes_k d}\right)^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}]$, and*

$$\left(\left(S^{\otimes_k d}\right)^{G^{d-1}}\right)^{S_d} \;\cong\; k[e_1, \ldots, e_d, e_d^{-1}][Y]/\left(Y^p - Y - \alpha\right).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$ of (2.6),

$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec}\widehat{K}_\eta} \;\cong\; \operatorname{Spec}\ \widehat{K}_\eta[Z]/\left(Z^p - Z - \alpha\right).$$

Proof. 1 Recall that $g \in G = \mathbb{Z}/p\mathbb{Z}$ acts on X by $g(X) = X + g$. Hence $(g_1, \ldots, g_{d-1}) \in K_G \cong G^{d-1}$ acts on the generators by

$$X_1 \mapsto X_1 + g_1, \qquad X_i \mapsto X_i - g_{i-1} + g_i \ (1 < i < d), \qquad X_d \mapsto X_d - g_{d-1},$$

so $Y = X_1 + \cdots + X_d$ changes by the telescoping sum $g_1 + (g_2 - g_1) + \cdots + (-g_{d-1}) = 0$: it is invariant. Moreover $\wp(Y) = \sum_i \wp(X_i) = \sum_i f(x_i) = \alpha$ ($\wp$ is additive). By Lemma 26 (with $r = 1$ and $A = R^{\otimes_k d}$), $T := R^{\otimes_k d}[Y]/(Y^p - Y - \alpha)$ is a $\mathbb{Z}/p\mathbb{Z}$-torsor over U^d, and $Y \mapsto X_1 + \cdots + X_d$ defines an $R^{\otimes_k d}$-algebra homomorphism $\phi : T \to \left(S^{\otimes_k d}\right)^{G^{d-1}}$. The target is the d-fold product torsor, a G-torsor over U^d by Proposition 25 and the surrounding discussion; its residual G-action is induced by $(g, 0, \ldots, 0) \in G^d$, under which $Y \mapsto Y + g$. So $\operatorname{Spec}\phi$ is a G-equivariant morphism of G-torsors over U^d, hence an isomorphism by Proposition 5.

2 S_d permutes the x_i and the X_i, so it fixes Y and α; in particular $\alpha \in \left(R^{\otimes_k d}\right)^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}]$. T is free over $R^{\otimes_k d}$ with basis the S_d-invariant monomials Y^j, $0 \le j < p$ (Lemma 26); writing an element in this basis, it is S_d-invariant if and only if all its coefficients are, whence

$$T^{S_d} = k[e_1, \ldots, e_d, e_d^{-1}][Y]/\left(Y^p - Y - \alpha\right).$$

The final statement follows from the scaffold exactly as in Proposition 44. ∎

Step 3: the degree count and the conclusion

Theorem 48 (the wild case, $G = \mathbb{Z}/p\mathbb{Z}$)**.** *Let $G = \mathbb{Z}/p\mathbb{Z}$ and let $\mathcal{P} \to U = C \setminus \mathfrak{m}$ be a G-torsor with ramification bounded by $\mathfrak{n} = nP$ at P. Then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Proof. Realize. The restriction $\mathcal{Q} = \mathcal{P}_x$ is a $\mathbb{Z}/p\mathbb{Z}$-torsor with ramification bounded by d, so Lemma 46 produces $\mathcal{P}' : X^p - X = f$ over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{P}_x$, where f is reduced with pole order $m < d$ (or $f = 0$). By Corollary 39, we may assume $\mathcal{P} = \mathcal{P}'$.

Identify. Proposition 47 identifies

$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \cong \operatorname{Spec} \widehat{K}_\eta[Z]/\big(Z^p - Z - \alpha\big), \qquad \alpha = \sum_{i=1}^{d} f(x_i).$$

Count. In the variables $y_i = x_i^{-1}$, each $f(x_i)$ is a polynomial in y_i of degree m with zero constant term, so α is a symmetric polynomial in $y_1, \dots, y_d$ all of whose monomials have total degree $\leq m < d$. By Lemma 42, $\alpha \in \kappa(\eta) \subset \widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$; in particular $v_{\widehat{K}_\eta}(\alpha) \geq 0$. By Theorem 16(1)–(2), the extension defined by $Z^p - Z = \alpha$ is trivial or unramified; either way, $\mathcal{P}'^{[d]}$ — and therefore $\mathcal{P}^{[d]}$ — is unramified at $\eta_{\mathfrak{n}}$. ☐

We record the two cyclic conclusions obtained so far in the form used later.

Corollary 49. *In the general settings of a curve C, and $\mathcal{P} \to U = C \setminus \mathfrak{m}$ a G-torsor with ramification bounded by $\mathfrak{n} = nP \subset \mathfrak{m}$ at P, we have:*

1. *If $G = \mathbb{Z}/p\mathbb{Z}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$*
2. *If $\mathcal{P}$ is tamely ramified at $\mathfrak{n}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$*

Proof. By Remark 37 we may assume k is algebraically closed, so that $P \in C(k)$ and $\deg \mathfrak{n} = n$. Part (1) is Theorem 48, and part (2) is Theorem 45. ☐

2.5.3 The Wild Case $G = \mathbb{Z}/p^r\mathbb{Z}$

Next, we generalize to the case $G = \mathbb{Z}/p^r\mathbb{Z}$.

Theorem 50. *Assume $G = \mathbb{Z}/p^r\mathbb{Z}$. Then $\mathcal{P}^{[\deg(\mathfrak{n})]}$ is unramified at $\eta_{\mathfrak{n}}$.*

The proof occupies the remainder of this subsection. In contrast with what the phrase "we generalize" may suggest, the argument is *not* an induction on r with Theorem 48 as its base case: it is a direct argument, treating all r at once, in which each of the three steps of Section 2.5.2 — realization of the local torsor by an explicit equation over $\mathbb{G}_m$, identification of the symmetric-power cover at $\eta_{\mathfrak{n}}$, and the degree estimate — is replaced by its Witt-vector analogue, using the theory collected in Section 2.3.3 and Section 2.3.4. In particular the case $r = 1$ is *reproved* (as the special case of Witt vectors of length one) rather than used. For an explanation of why the seemingly natural induction on r fails — in short: the hypothesis bounds the ramification of $\mathcal{P}$ in *upper* numbering, while the break of the top degree-p layer $\mathcal{P} \to \mathcal{P}'$ is the largest *lower*-numbering break, which can be of size about $p^{r-1}d$ — see Chapter A.

Standing reduction

As in the preceding cases, we may assume k is algebraically closed (Remark 37). We may further assume that $\mathcal{P} \to U$ is *totally ramified* at P with inertia equal to all of G: writing $R = \widehat{\mathcal{O}_{C,P}}$, $K = \mathrm{Frac}(R)$, and letting $H = \rho(G^0) \subseteq G$ be the image of the inertia subgroup under the homomorphism $\rho\colon \mathrm{Gal}(K^{sep}/K) \to G$ classifying $\mathcal{P}|_{\operatorname{Spec} K}$, decompose $\mathcal{P} \to \mathcal{P}/H \to U$ as in Proposition 15, where $\mathcal{P} \to \mathcal{P}/H$ is an H-torsor and $\mathcal{P}/H \to U$ a G/H-torsor. Then $\mathcal{P}/H$ is unramified at P, hence extends over P; taking any point $Q \in \mathcal{P}/H$ over P yields $\widehat{\mathcal{O}_{\mathcal{P}/H,Q}} \cong \widehat{\mathcal{O}_{C,P}}$, and since k is algebraically closed the torsor $\mathcal{P}/H$ is locally trivial at Q. This replaces $(G, \mathcal{P} \to U)$ by $(H, \mathcal{P} \to \mathcal{P}/H)$ with $H \cong \mathbb{Z}/p^s\mathbb{Z}$ for some $s \leq r$; as the theorem is proved for all r simultaneously, we may rename and assume $H = G$.

After this reduction, the restriction $\mathcal{Q} := \mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc $\operatorname{Spec} k((x))$ at P (with x a uniformizer at P) is a *connected* $\mathbb{Z}/p^r\mathbb{Z}$-torsor: $\mathcal{Q} = \operatorname{Spec} L$ for a cyclic, totally ramified extension L/K of degree p^r, with ramification bounded by $d := \deg \mathfrak{n}$ (Definition 7); equivalently, the largest upper-numbering break u of L/K satisfies $u < d$ *strictly.*

Step 1: realization over $\mathbb{G}_m$

The first step generalizes Lemma 46 from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{Z}/p^r\mathbb{Z}$, using the reduced polynomial representatives of Section 2.3.4.

Lemma 51 (Realization over $\mathbb{G}_m$ for W_r)**.** *Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{Q}$ be a* connected *$\mathbb{Z}/p^r\mathbb{Z}$-torsor over* $\operatorname{Spec}(k((x)))$ *with ramification bounded by d (Definition 7). Then there is a reduced polynomial vector $\vec{f} = (f_0, \ldots, f_{r-1})$, $f_j \in x^{-1}k[x^{-1}]$, such that, with $m_j := \deg_{x^{-1}} f_j$:*

1. *$p^{r-1-j}\, m_j < d$ for all $0 \leq j \leq r-1$; equivalently $m_j < d\, p^{j-(r-1)}$;*
2. *$\mathcal{P}' := \operatorname{Spec}\Big(k[x, x^{-1}][\vec{X}]/\big(\wp\vec{X} \ominus \vec{f}\big)\Big)$ is a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$;*
3. *$\mathcal{P}'$ extends to a torsor over $\mathbb{P}^1_k \setminus \{0\}$ — in particular it is unramified at ∞ — and has ramification bounded by d at 0.*

Proof. $\mathcal{Q} = \operatorname{Spec} L$ with L/K cyclic totally ramified of degree p^r and largest upper break $u < d$ (Theorem 27). Let $[\vec{f_0}] \in W_r(K)/\wp$ be its Artin–Schreier–Witt class and let $\vec{f}$ be a reduced polynomial representative (Proposition 34).

First, $f_0 \neq 0$: otherwise Corollary 28 would bound the order of the class by p^{r-1}, contradicting $[L : K] = p^r$. Hence $m_0 \geq 1$ and $p \nmid m_0$, and the Schmid–Witt break formula (Theorem 31) applies: $\max_j p^{r-1-j} m_j = u < d$, which is 1.

For 2: by Lemma 26 (with $A = k[x, x^{-1}] \ni f_j$), $\mathcal{P}'$ is a $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\mathbb{G}_m$; its restriction to $\operatorname{Spec} k((x))$ is the Artin–Schreier–Witt torsor of the class of $\vec{f}$, i.e. $\mathcal{Q}$ (Theorem 27).

For 3: in the coordinate $y = x^{-1}$ at ∞ we have $f_j \in y\,k[y]$, so $\vec{f} \in W_r(k[y])$ and Lemma 26 (with $A = k[y]$) extends $\mathcal{P}'$ to a torsor over $\mathbb{A}^1_k = \operatorname{Spec} k[y] = \mathbb{P}^1_k \setminus \{0\}$; a torsor over the full local ring at ∞ is unramified there. The ramification bound at 0 holds because $\mathcal{P}'_x \cong \mathcal{Q}$. ☐

Remark. Connectedness of $\mathcal{Q}$ is guaranteed in the application by the standing reduction (inertia equal to all of G). Note also the following sanity check: for every j with $p^{r-1-j} \geq d$, the bound 1 forces $m_j = 0$, i.e. $f_j = 0$ — all sufficiently low Witt components of a reduced representative vanish outright. This makes visible how strong the hypothesis $u < d$ is at the lower levels.

Step 2: identification of the symmetric-power cover

We now generalize the invariant-ring computation of Proposition 47 to Witt coordinates. Fix $\vec{f} \in W_r(R)$, $R = k[x, x^{-1}]$, and let

$$S \;=\; R[\vec{X}]/\big(\wp\vec{X} \ominus \vec{f}\big)$$

be the coordinate ring of the torsor $\mathcal{P}' \to \mathbb{G}_m$ of Lemma 51. Write $R^{\otimes_k d} = k[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$ and

$$S^{\otimes_k d} = R^{\otimes_k d}[\vec{X}_1, \dots, \vec{X}_d]/\big(\wp\vec{X}_1 \ominus \vec{f}(x_1), \dots, \wp\vec{X}_d \ominus \vec{f}(x_d)\big),$$

where $\vec{X}_i = (X_{i,0}, \dots, X_{i,r-1})$ and $\vec{f}(x_i)$ is the image of $\vec{f}$ under $x \mapsto x_i$. Exactly as in the scaffold of Section 2.4, the d-fold product torsor $p_1^{-1}\mathcal{P}' \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P}'$ on U^d is the quotient of $\operatorname{Spec} S^{\otimes_k d}$ by $K_G \cong G^{d-1}$, embedded in G^d as the kernel of the sum map via $(g_1, \dots, g_{d-1}) \mapsto (g_1, g_2 - g_1, \dots, -g_{d-1})$, and acting (in Witt coordinates, writing $\underline{g}$ for the image of $g \in G$ under $\mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p)$, Example 23) by

$$(g_1, \dots, g_{d-1})\colon \quad \vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}_1, \qquad \vec{X}_i \mapsto \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1} \; (1 < i < d), \qquad \vec{X}_d \mapsto \vec{X}_d \ominus \underline{g}_{d-1}.$$

Proposition 52 (identification of the product torsor, Witt case). *Set*

$$\vec{Y} := \bigoplus_{i=1}^{d} \vec{X}_i \in W_r\big(S^{\otimes_k d}\big), \qquad \vec{\alpha} := \bigoplus_{i=1}^{d} \vec{f}(x_i) \in W_r\big(R^{\otimes_k d}\big),$$

where $\bigoplus$ denotes the d-fold Witt vector sum $\oplus$ (not a direct sum); in particular $\vec{Y}$ is a single Witt vector of length r. Then:

1. *Every component Y_n of $\vec{Y}$ is G^{d-1}-invariant, and*

$$\big(S^{\otimes_k d}\big)^{G^{d-1}} \;\cong\; R^{\otimes_k d}[\vec{Y}]/\big(\wp\vec{Y} \ominus \vec{\alpha}\big)$$

   as G-torsors over $R^{\otimes_k d}$ (the residual $G = G^d/G^{d-1}$ acting by $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$).

2. *Every component α_n of $\vec{\alpha}$ is a symmetric Laurent polynomial, i.e. lies in $\big(R^{\otimes_k d}\big)^{S_d} = k[e_1, \dots, e_d, e_d^{-1}]$, and*

$$\Big(\big(S^{\otimes_k d}\big)^{G^{d-1}}\Big)^{S_d} \;\cong\; k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/\big(\wp\vec{Y} \ominus \vec{\alpha}\big).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$, $\kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$, of (2.6),

$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \;\cong\; \operatorname{Spec} \; \widehat{K}_\eta[\vec{Z}]/\big(\wp\vec{Z} \ominus \vec{\alpha}\big).$$

Proof. 1 An element $\sigma = (g_1, \dots, g_{d-1}) \in G^{d-1}$ acts as a ring automorphism of $S^{\otimes_k d}$: this is its *original* action, defined on the generators $X_{i,n}$ by the twisted formulas above (which mix components through the carry polynomials of (2.3)). By functoriality of W_r (Section 2.3.3), the induced map $W_r(\sigma)$ on $W_r(S^{\otimes_k d})$ is componentwise and commutes with $\oplus, \ominus$; on the generators it reproduces the twisted action, $W_r(\sigma)(\vec{X}_i) = \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1}$ (with $\underline{g}_0 = \underline{g}_d = \vec{0}$). Hence, using commutativity and associativity of $\oplus$, the Witt sum transforms by the telescoping constant

$$W_r(\sigma)(\vec{Y}) \;=\; \bigoplus_{i=1}^{d} W_r(\sigma)(\vec{X}_i) \;=\; \vec{Y} \oplus \underline{g}_1 \oplus (\underline{g}_2 \ominus \underline{g}_1) \oplus \cdots \oplus (\ominus\, \underline{g}_{d-1}) \;=\; \vec{Y}.$$

Since $W_r(\sigma)$ is componentwise, this equality of r-tuples says exactly $\sigma(Y_n) = Y_n$ for every n, i.e. each $Y_n \in S^{\otimes_k d}$ is invariant. Since $\wp = F - \mathrm{id}$ is additive (Definition 24 ff.),

$$\wp\vec{Y} = \bigoplus_{i=1}^{d} \wp\vec{X}_i = \bigoplus_{i=1}^{d} \vec{f}(x_i) = \vec{\alpha}.$$

Hence there is a well-defined $R^{\otimes_k d}$-algebra homomorphism

$$\phi\colon\; T := R^{\otimes_k d}[\vec{Z}]/\big(\wp\vec{Z} \ominus \vec{\alpha}\big) \;\longrightarrow\; \big(S^{\otimes_k d}\big)^{G^{d-1}}, \qquad \vec{Z} \mapsto \vec{Y}.$$

By Lemma 26, $\operatorname{Spec} T$ is a G-torsor over U^d. The target is the product torsor (the contracted product of the $p_i^{-1}\mathcal{P}'$ along the sum map $G^d \to G$), a G-torsor over U^d by Proposition 25 and the surrounding discussion; its residual G-action is induced by $(g, 0, \dots, 0) \in G^d$, i.e. $\vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}$, under which $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$. So $\operatorname{Spec}\phi$ is a G-equivariant morphism of G-torsors over U^d, hence an isomorphism by Proposition 5.

2 S_d acts on $S^{\otimes_k d}$ by permuting the tensor factors, i.e. $x_i \mapsto x_{\sigma(i)}$, $\vec{X}_i \mapsto \vec{X}_{\sigma(i)}$. As in 1, each $\sigma \in S_d$ is a ring automorphism, so $W_r(\sigma)$ is componentwise and commutes with $\oplus$. Since $\oplus$ is commutative and associative, permuting the summands of a Witt sum leaves it unchanged; thus $\vec{Y}$ is S_d-invariant (componentwise, as before) and $W_r(\sigma)(\vec{\alpha}) = \bigoplus_i \vec{f}(x_{\sigma(i)}) = \vec{\alpha}$, so each $\alpha_n \in R^{\otimes_k d}$ is a symmetric Laurent polynomial, hence lies in $k[e_1, \dots, e_d, e_d^{-1}]$ (the S_d-invariants of $R^{\otimes_k d}$, as recorded in the scaffold). By Lemma 26, T is free over $R^{\otimes_k d}$ with basis the monomials $\prod_n Y_n^{a_n}$, $0 \le a_n < p$, each of which is S_d-invariant; writing an element in this basis, it is S_d-invariant if and only if all its coefficients are. Therefore

$$T^{S_d} = \big(R^{\otimes_k d}\big)^{S_d}[\vec{Y}]/\big(\wp\vec{Y} \ominus \vec{\alpha}\big) = k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/\big(\wp\vec{Y} \ominus \vec{\alpha}\big).$$

The final statement follows exactly as in the $r = 1$ case: by definition $\mathcal{P}'^{[d]}$ is the quotient by $\Xi = K_G \rtimes S_d$ computed above (Corollary 27), and its restriction to $\operatorname{Spec} \widehat{K}_\eta$ along $k[e_1, \dots, e_d, e_d^{-1}] \to \widehat{K}_\eta$ is the Artin–Schreier–Witt cover of the image of $\vec{\alpha}$ in $W_r(\widehat{K}_\eta)$. ☐

Remark. A pure restriction argument on H^1's ("the class of the product torsor is the sum of the pulled-back characters") identifies the cover over $k(x_1, \dots, x_d)$ but does *not* by itself descend it to the symmetric base $k(e_1, \dots, e_d)$; the invariant-theoretic computation above — like its Kummer and Artin–Schreier counterparts, Propositions 44 and 47 — is what effects the descent.

Step 3: the degree count

Throughout this paragraph, $\vec{f}$ is the reduced polynomial vector of Lemma 51, so $m_j = \deg_{x^{-1}} f_j$ satisfies $m_j < d\,p^{j-(r-1)}$, and $y_i := 1/x_i$, so that $f_j(x_i) \in y_i\, k[y_i]$ is a polynomial in y_i of degree m_j.

Theorem 53 (regularity of all Witt components). *With $\vec{f}$ as in Lemma 51, every component α_n $(0 \le n \le r-1)$ of $\vec{\alpha} = \bigoplus_{i=1}^{d} \vec{f}(x_i)$ lies in $\kappa(\eta)$; in particular $\vec{\alpha} \in W_r\big(\widehat{\mathcal{O}}_\eta\big)$.*

Proof. Fix $n \le r-1$ and a monomial $\prod_{i,j}\big(f_j(x_i)\big)^{e_{ij}}$ of α_n, viewed as a universal isobaric polynomial in the $f_j(x_i)$ of weight p^n (Lemma 29, with $f_j(x_i)$ of weight p^j); set $E_j := \sum_i e_{ij}$, so $\sum_j E_j p^j = p^n$. Expanding each $f_j(x_i)$ as a polynomial in y_i of degree m_j, every resulting monomial in the y_i has total degree

$$\le \sum_{i,j} e_{ij}\, m_j \;=\; \sum_j E_j\, m_j \;<\; \sum_j E_j\, d\, p^{\,j-(r-1)} \;=\; d\, p^{-(r-1)} \sum_j E_j p^j \;=\; d\, p^{\,n-(r-1)} \;\le\; d,$$

strictly: some $E_j > 0$ (as $\sum_j E_j p^j = p^n > 0$), and the pole bound $m_j < d\, p^{\,j-(r-1)}$ of Lemma 511 is strict even when $m_j = 0$. Thus every monomial of α_n — a symmetric polynomial in $y_1, \ldots, y_d$ by Proposition 522 — has total y-degree $< d$, and Lemma 42 gives $\alpha_n \in \kappa(\eta)$. ☐

Remark. The bound is exact: the weight $p^{\,j-(r-1)}$ is paired against the isobaricity constraint $\sum_j E_j p^j = p^n$, so the estimate $d\, p^{\,n-(r-1)} \le d$ is independent of how the Witt carries distribute the exponents E_j; the hypothesis "break $< d$ at P" is used at *every* Witt level at once. The sanity check of Lemma 51 records the extreme case: components with $p^{\,r-1-j} \ge d$ vanish outright in the reduced representative.

Proof of the theorem

Proof of Theorem 50. By the standing reduction above we may assume k algebraically closed and $\mathcal{P} \to U$ totally ramified at P with inertia all of G, so that $\mathcal{Q} := \mathcal{P}_x$ is a connected $\mathbb{Z}/p^r\mathbb{Z}$-torsor over $\operatorname{Spec} k((x))$ with ramification bounded by $d = \deg \mathfrak{n}$.

Lemma 51 produces a torsor $\mathcal{P}' \to \mathbb{G}_m$, defined by a reduced polynomial vector $\vec{f}$ with $p^{r-1-j} m_j < d$, with $\mathcal{P}'_x \cong \mathcal{Q}$. By Corollary 39 (valid for arbitrary finite abelian G), the ramification of $\mathcal{P}^{[d]}$ at $\eta_\mathfrak{n}$ agrees with that of $\mathcal{P}'^{[d]}$ at $\eta_{d\cdot 0}$; so we may assume $C = \mathbb{P}^1_k$, $\mathfrak{m} = d \cdot 0 + \infty$, $\mathfrak{n} = d \cdot 0$, $\mathcal{P} = \mathcal{P}'$.

By Proposition 52,

$$\mathcal{P}'^{[d]}\big|_{\operatorname{Spec} \widehat{K}_\eta} \;\cong\; \operatorname{Spec}\; \widehat{K}_\eta[\vec{Z}]/\big(\wp \vec{Z} \ominus \vec{\alpha}\big), \qquad \vec{\alpha} = \bigoplus_{i=1}^{d} \vec{f}(x_i),$$

with $\widehat{K}_\eta = \kappa(\eta)((e_d))$ the complete local field at $\eta_\mathfrak{n}$ ((2.6)) and $\widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$ its valuation ring.

By Theorem 53, $\vec{\alpha} \in W_r(\kappa(\eta)) \subset W_r(\widehat{\mathcal{O}}_\eta)$. Hence, by Lemma 26 applied over $A = \widehat{\mathcal{O}}_\eta$, the Artin–Schreier–Witt cover $\wp\vec{Z} = \vec{\alpha}$ is finite étale over $\widehat{\mathcal{O}}_\eta$, and $\mathcal{P}'^{[d]}|_{\operatorname{Spec} \widehat{K}_\eta}$ is its generic fibre: every field factor of $\widehat{K}_\eta[\vec{Z}]/(\wp\vec{Z} \ominus \vec{\alpha})$ is the fraction field of a finite étale $\widehat{\mathcal{O}}_\eta$-algebra, i.e. an unramified extension of $\widehat{K}_\eta$ with respect to the discrete valuation ring $\widehat{\mathcal{O}}_\eta$. By Definition 6, $\mathcal{P}^{[d]}$ is unramified at $\eta_\mathfrak{n}$. ☐

Remark. The argument proves all r simultaneously; it reproves the case $r = 1$ (Theorem 48) as the special case of a vector of length one and never invokes it.

2.5.4 The General Finite Abelian Case

We now assemble the general case from the three cyclic computations. The mechanism is the structure theorem for finite abelian groups together with the compatibility of the symmetric power with fiber products (Lemma 40).

Proof of Theorem 1. Fix a single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, write $P = P_i$ and $d = d_{\mathfrak{n}}$. By Remark 37 we may assume k is algebraically closed, so $d = n_i$.

By the structure theorem for finite abelian groups, fix an isomorphism

$$G \;\cong\; \prod_{j=1}^{s} \mathbb{Z}/q_j\mathbb{Z}, \qquad q_j = \ell_j^{r_j} \text{ prime powers,}$$

and let $\pi_j : G \to \mathbb{Z}/q_j\mathbb{Z}$ be the projections. Let $\mathcal{P}_j := \mathcal{P}/\ker(\pi_j)$ be the quotient torsor (Proposition 15); it is a $\mathbb{Z}/q_j\mathbb{Z}$-torsor over U, and at each point of $\mathrm{Supp}(\mathfrak{m})$ its classifying character is $\pi_j \circ \rho$, where ρ is that of $\mathcal{P}$.

Step 1: $\mathcal{P}$ is the fiber product of its primary components. The quotient maps $\mathcal{P} \to \mathcal{P}_j$ combine to a morphism

$$\mathcal{P} \longrightarrow \mathcal{P}_1 \times_U \mathcal{P}_2 \times_U \cdots \times_U \mathcal{P}_s$$

which is equivariant for the isomorphism $G \cong \prod_j \mathbb{Z}/q_j\mathbb{Z}$. The target, being a fiber product of $\mathbb{Z}/q_j\mathbb{Z}$-torsors, is a $\prod_j \mathbb{Z}/q_j\mathbb{Z}$-torsor, so the morphism is an isomorphism of G-torsors (Proposition 5).

Step 2: each component satisfies the hypothesis of one of the three cyclic cases. Since $\rho(G^{n_i}_{\widehat{K}_{P_i}}) = \{1\}$ implies $(\pi_j \circ \rho)(G^{n_i}_{\widehat{K}_{P_i}}) = \{1\}$, each $\mathcal{P}_j$ has ramification bounded by $\mathfrak{m}$; in particular, bounded by $\mathfrak{n} = dP$ at P. Moreover:

- If $\ell_j \neq p$: the image of the wild inertia subgroup under $\pi_j \circ \rho$ is a p-group inside $\mathbb{Z}/q_j\mathbb{Z}$, whose order is prime to p; so it is trivial, and $\mathcal{P}_j$ is *tamely ramified* at every point of $\mathrm{Supp}(\mathfrak{m})$ — in particular at P.
- If $\ell_j = p$: $\mathcal{P}_j$ is a $\mathbb{Z}/p^{r_j}\mathbb{Z}$-torsor with ramification bounded by $\mathfrak{n}$ at P.

Step 3: the cyclic cases. By Theorem 45, for every j with $\ell_j \neq p$ the torsor $\mathcal{P}_j^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$. By Theorem 50, for every j with $\ell_j = p$ the torsor $\mathcal{P}_j^{[d]}$ is unramified — in particular tamely ramified — at $\eta_{\mathfrak{n}}$.

Step 4: reassembly. Applying Lemma 40 repeatedly to Step 1,

$$\mathcal{P}^{[d]} \;\cong\; \mathcal{P}_1^{[d]} \times_{U^{(d)}} \mathcal{P}_2^{[d]} \times_{U^{(d)}} \cdots \times_{U^{(d)}} \mathcal{P}_s^{[d]}.$$

By Lemma 12 (applied repeatedly at the prime divisor $(E_{\mathfrak{n}}, \eta_{\mathfrak{n}})$ of $X_{\mathfrak{n}}$), a fiber product of tamely ramified torsors is tamely ramified. Hence $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$, as claimed. $\square$

Remark. The proof shows slightly more: the wild part of $\mathcal{P}$ contributes *unramified* factors at $\eta_{\mathfrak{n}}$, so the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is entirely accounted for by the tame, prime-to-p part of $\mathcal{P}$ — with ramification index dividing the order of the tame quotient of the local character at P, as computed in Theorem 45.

2.6 Ramification on Products of Blowups

We conclude this chapter by proving some auxiliary results that will be useful in Chapter 3.

Let X and Y be smooth schemes over a field k, and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \mathrm{Bl}_x(X) \to X$ and $\pi_Y : \mathrm{Bl}_y(Y) \to Y$.

Furthermore, let $\pi_{X\times Y} : \mathrm{Bl}_{(x,y)}(X \times_k Y) \to X \times_k Y$ be the blowup of the product scheme at the point (x, y). We denote by E_X, E_Y, and $E_{X\times Y}$ the respective exceptional divisors, and let η_X, η_Y, and $\eta_{X\times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 54. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G-torsors defined on open subsets $U_X \subset \mathrm{Bl}_x(X)$ and $U_Y \subset \mathrm{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r, then the external product torsor*

$$\mathcal{G}_{X\times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor in the product blowup.

The proposition is purely local in nature; it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y, we may restrict our attention to open neighborhoods of x and y that are étale over affine spaces. The rest of this section treats that case. So let $X = \mathbb{A}_k^n$ and $Y = \mathbb{A}_k^m$ with origins $x = 0$ and $y = 0$; the local rings, residue fields and valuations at the generic points η_X and η_Y of the exceptional divisors $E_X \subset \mathrm{Bl}_0(X)$ and $E_Y \subset \mathrm{Bl}_0(Y)$ are the ones computed in Proposition 35.

The Product Case

As before, $\mathrm{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\mathrm{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0, 0)$, denoted $\mathrm{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \le i, j \le n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \le k, l \le m \\ x_i w_{n+j} = y_j w_i & 1 \le i \le n,\ 1 \le j \le m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X\times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$. They share a common dense open set $\tilde{U}$ defined by the condition that neither the X-coordinates nor the Y-coordinates vanish simultaneously in the projective space:

$$\tilde{U} = \{((x, y), [w_1 : \cdots : w_{n+m}]) \mid (w_1, \ldots, w_n) \neq 0 \text{ and } (w_{n+1}, \ldots, w_{n+m}) \neq 0\}$$

This yields a diagram of open immersions:

$$\begin{tikzcd} & \tilde{U} \arrow[dl, "f_1"'] \arrow[dr, hook, "f_2"] & \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & \mathrm{Bl}_{(0,0)}(X \times_k Y) \end{tikzcd}$$

where f_1 maps the coordinates to the respective projectivizations $[w_1 : \cdots : w_n]$ and $[w_{n+1} : \cdots : w_{n+m}]$. Also note that the generic point $\eta_{X\times Y}$ of $E_{X\times_k Y}$ is inside $\tilde{U}$.

Extensions of DVRs

Let S be the local ring of the generic point $\eta_{X\times Y}$ of $E_{X\times Y}$ in $\tilde{U}$. On the chart where $w_1 \neq 0$ and $w_{n+1} \neq 0$, we have $x_1 = (\frac{w_1}{w_{n+1}})y_1$. Since $\frac{w_1}{w_{n+1}}$ is a unit in this chart, x_1 and y_1 are equivalent as uniformizers. We have:

$$S = k\left[x_1, \frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right]_{(x_1)} = k\left[\frac{w_1}{w_{n+1}}, \frac{w_2}{w_{n+1}} \cdots \frac{w_n}{w_{n+1}}, y_1 \cdots \frac{w_{n+m}}{w_{n+1}}\right]_{(y_1)}$$

$$k(\eta_{X\times Y}) = k\left(\frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1}\right)$$

Let R_X be the local ring of the exceptional divisor E_X in $\mathrm{Bl}_0(X)$ (computed in Proposition 35). The pullback of $E_X \times_k \mathrm{Bl}_0(Y)$ along f_1 induces an extension of DVRs $R_X \hookrightarrow S$. Which is:

1. Weakly Unramified: x_1 is a uniformizer in both R_X and S, so the ramification index is $e = 1$.
2. Residually Transcendental: The residue field extension $\kappa(\eta_X) \subset \kappa(\eta)$ is:

$$k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}\right) \subset k\left(\frac{w_2}{w_1}, \ldots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1}, \ldots, \frac{w_{n+m}}{w_1}\right)$$

   Hence separable.

Since this extension is generated by transcendental elements, it is separable and formally smooth at the maximal ideal ([Stacks, Tag 09E7]).

Ramification of G-Torsors on the Product

Let G be a finite abelian group. Let $\mathcal{Q}$ be a G-torsor on an open $U \subset \mathrm{Bl}_0(X)$ disjoint from E_X. Let $V \subset \mathrm{Bl}_0(Y)$ be an open subscheme, and let $\pi_X : \mathrm{Bl}_0(X) \times_k V \to \mathrm{Bl}_0(X)$ be the projection onto the first factor. By restricting this projection to $U \times_k V$, we obtain the pullback G-torsor:

$$\pi_X^{-1}(\mathcal{Q}) \cong \mathcal{Q} \times_k V$$

which is defined on the open subset $U \times_k V \subset \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$. The extension of local rings $R_X \to S$ is weakly unramified (the ramification index $e = 1$) and residually transcendental with separable residue field extension. Under these conditions the ramification filtration is preserved. Therefore, the pullback $\pi_X^{-1}(\mathcal{Q})$ has ramification bounded by r at the generic point $\eta_{X\times Y}$ of the exceptional divisor $E_{X\times Y}$ if and only if the original torsor $\mathcal{Q}$ has ramification bounded by r at the generic point η_X of E_X.

And we finish by Lemma 10.

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